

ment to reduce noise. The APU firewall is of minimum size, and the APU is easily accessible from a workstand.

Military transports with the landing gear in fuselage-mounted pods can place the APU in the pod. This provides ground-level access to the APU, but requires increased firewall area.

Fighters usually have the APU in the fuselage, near the hydraulic pumps and generators. This requires a firewall that completely encloses the APU.

APU installation is discussed in detail in Ref. 34.

Avionics

Avionics (a contraction of “aviation electronics”) includes radios, flight instruments, navigational aids, flight control computers, radar, and other aircraft sensors such as infrared detectors. For initial layout, it is necessary to provide sufficient volume in the avionics bays. Also, the nose of the aircraft should be designed to hold the radar.

On the average, avionics has a density of about 30–45 lb/ft³. The required avionics weight can be estimated from the aircraft empty weight (W_e), which is known at this point. Table 11.6 provides ratios between avionics weight and aircraft empty weight.

Estimation of radar size is very complex, and depends upon the desired detection range, radar cross section of the threat aircraft, and radar frequency. For initial design, the radar from a similar design should be used until the avionics group provides a definition of the required radar.

In the absence of better information, it can be assumed that a bomber will use a 40-in. radar, a large fighter will use a 35-in. radar, and a small fighter will use a 22-in. radar. Transport-aircraft radars are only for weather avoidance. They are small relative to the size of the aircraft nose and can be initially ignored.

12.1 INTRODUCTION

The previous chapters have presented methods for the design layout of a credible aircraft configuration. Initial sizing, wing geometry, engine installation, tail geometry, fuselage internal arrangement, and numerous other design topics have been discussed.

The initial sizing was based upon rough estimates of the aircraft’s aerodynamics, weights, and propulsion characteristics. At that time we could not calculate the actual characteristics of the design because the aircraft had not been designed yet!

Now the aircraft design can be analyzed “as-drawn” to see if it actually meets the required mission range. If not, we will resize the aircraft until it does.

Also, a variety of trade studies can now be performed to determine the best combination of design parameters (T/W , W/S , aspect ratio, etc.) to meet the given mission and performance requirements at the minimum weight and cost.

Additional analysis on the as-drawn aircraft is also required at this time to insure that stability and control requirements are met. In previous chapters, an approximate tail volume coefficient method was used for tail sizing. Now that the aircraft is drawn, we can analytically determine if the selected tail sizes are adequate.

These methods will be presented in Chapters 12–19. The overall objective of these chapters is the understanding of the sizing optimization and trade study process, not the presentation of particular theories and analysis methods. Presumably the student has spent several years studying aerodynamics, controls, structures, and propulsion.

The analysis techniques presented in these chapters, all approximate methods, illustrate the major parameters to be determined and provide realistic trends for trade studies. In many cases these are not the methods employed by the major aircraft companies, whose methods are highly computerized and cannot be presented in any single textbook. Also, each company uses many proprietary methods that are simply unavailable to students.

By using the simplified methods provided in this book, the student designer will experience the interaction of the major design variables but will not devote excessive time to the analytical tasks. This should leave more time for learning the basic principles of sizing optimization and trade studies, which are the same regardless of analytical techniques.

12.2 AERODYNAMIC FORCES

Figure 12.1 shows the only two ways that the airmass and the airplane can act upon each other. As the aircraft moves forward, the air molecules slide over its skin. The molecules closest to the skin act as if they are stuck to it, moving with the aircraft (“no-slip condition”).

If the air molecules closest to the aircraft skin are moving with it, there must be slippage (or “shear”) between these molecules and the nonmoving molecules away from the aircraft. “Viscosity” is the honey-like tendency of air to resist shear deformation, which causes additional air near the aircraft skin to be dragged along with the aircraft. The force required to accelerate this “boundary-layer” air in the direction the aircraft is travelling produces skin-friction drag.

If the air molecules slide over each other (shear) in an orderly fashion, the flow is said to be “laminar.” If the molecules shear in a disorderly fashion the flow is “turbulent.” This produces a thicker boundary layer, indicating that more air molecules are dragged along with the aircraft, generating more skin-friction drag.

Airflow along a smooth plate becomes turbulent when the local Reynolds number reaches about one-half million, but can become turbulent at a lower Reynolds number if there is substantial skin roughness. Also, the curvature of the surface can either prevent or encourage the transition from laminar to turbulent flow.

As the aircraft moves forward, the air molecules are pushed aside. This causes the relative velocity of the air to vary about the aircraft. In some places, mostly toward the nose, the air is slowed down. In other places the air is speeded up relative to the freestream velocity.

According to Bernoulli’s equation, the total pressure (static plus dynamic) along a subsonic streamline remains constant. If the local air velocity increases, the dynamic pressure has increased so the static pressure must decrease. Similarly, a reduction in local air velocity leads to an increase in static pressure.

Thus, the passage of the aircraft creates varying pressures around it, which push on the skin as shown in Fig. 12.1.

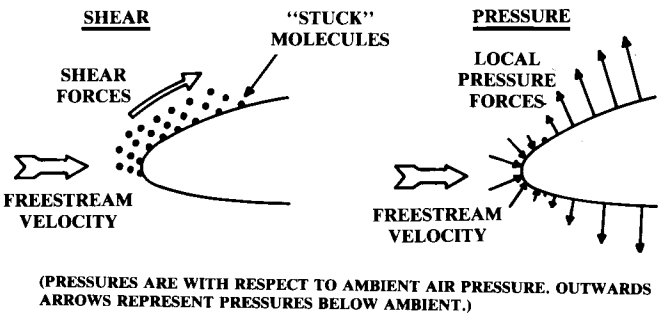


Fig. 12.1 Origin of aerodynamic forces.

In fact, lift is created by forcing the air that travels over the top of the wing to travel faster than the air which passes under it. This is accomplished by the wing’s angle of attack and/or wing camber. The resulting difference in air velocity creates a pressure differential between the upper and lower surfaces of the wing, which produces the lift that supports the aircraft.

If the aircraft is traveling near or above the speed of sound, additional pressure forces are produced by the shock waves around the aircraft. Shock waves result whenever supersonic flow is being slowed down.

All aerodynamic lift and drag forces result from the combination of shear and pressure forces. However, the dozens of classification schemes for aerodynamic forces can create considerable confusion because of overlapping terminology.

For example, the drag on a wing includes forces variously called airfoil profile drag, skin-friction drag, separation drag, parasite drag, camber drag, drag-due-to-lift, wave drag, wave-drag-due-to-lift, interference drag—and so forth.

Figure 12.2 presents the various drag terminologies using a matrix that defines the drag type based upon the origin of the drag force (shear or pressure) and whether or not the drag is strongly related to the lift force being developed.

Drag forces not strongly related to lift are usually known as parasite drag or zero-lift drag. In subsonic cruising flight of a well-designed aircraft, the

	SHEAR FORCES	PRESSURE FORCES		
		SEPARATION	SHOCK	CIRCULATION
PARASITE DRAG	SKIN FRICTION	VISCOUS SEPARATION	WAVE DRAG	
	SCRUBBING DRAG	SHOCK-INDUCED SEPARATION “DRAG RISE”		
	INTERFERENCE DRAG			
	PROFILE DRAG			
INDUCED DRAG [F(Lift)]	SUPERVELOCITY EFFECT ON SKIN FRICTION	CAMBER DRAG		DRAG DUE TO LIFT TRIM DRAG
		SUPERVELOCITY EFFECT ON PROFILE DRAG – i.e., LANDING GEAR, ETC.		
			WAVE DRAG DUE TO LIFT	
REFERENCE AREA:	S_{wetted}	Max. Cross Section	(Volume Distribution)	S_{ref}

Fig. 12.2 Drag terminology matrix.

parasite drag consists mostly of skin-friction drag, which depends mostly upon the wetted area.

The skin-friction drag of a flat plate of the same wetted area as the aircraft can be determined for various Reynolds numbers and skin roughnesses using equations provided below. However, the actual parasite drag will be somewhat larger than this value, as will be shown later.

“Scrubbing drag” is an increase in the skin-friction drag due to the propwash or jet exhaust impinging upon the aircraft skin. This produces a higher effective air velocity and assures turbulent flow, both of which increase drag. It is for this reason that pusher-propellers are desirable, and that few modern jets have conformal nacelles in which the exhaust rubs along the aft fuselage.

There are three separate origins of the drag-producing pressure forces. The first, viscous separation, was the source of considerable difficulty during the early theoretical development of aerodynamics.

If the theoretical pressure forces in a perfect fluid are integrated over a streamlined body without flow separation, it is found that the pressures around the body which yield a drag force in the flight direction are exactly matched by the pressures around the body which yield a forward force. Thus, if skin friction is ignored the net drag is zero.

This was known to be false, and was called d'Alembert's paradox. The paradox was finally resolved by Prandtl who determined that the boundary layer, which is produced by viscosity, causes the flow to separate somewhere on the back half of the body. This prevents the full attainment of the forward-acting force, leaving a net drag force due to viscous separation. (See Ref. 35 for a more detailed discussion.)

Viscous separation drag, also called “form drag,” depends upon the location of the separation point on the body. If the flow separates nearer to the front of the body the drag is much higher than if it separates more towards the rear.

The location of the separation point depends largely upon the curvature of the body. Also, the separation point is affected by the amount of energy in the flow. Turbulent air has more energy than laminar air, so a turbulent boundary layer actually tends to delay separation.

If a body is small and flying at low speed, the Reynolds number will be so low that the flow will remain laminar resulting in separated flow. For this reason a small body may actually have a lower total drag when its skin is rough. This produces turbulent flow, which will remain attached longer than would laminar flow. The dimples on a golf ball are an example of this.

For a very long body such as the fuselage of an airliner, the turbulent boundary layer will become so thick that the air near the skin loses most of its energy. This causes separation near the tail of the aircraft, resulting in high “boattail drag.”

To prevent this, small vanes perpendicular to the skin and angled to the airflow are placed just upstream of the separation point. These vanes produce vortices off their ends, which mix the boundary layer with higher-energy air from outside the boundary layer. This delays separation and reduces boattail drag. Such “vortex generators” are also used on wing and tail surfaces.

Viscous separation is largely responsible for the drag of irregular bodies such as landing gear and boundary-layer diverters. It also produces base drag, the pressure drag created by a “cut-off” aft fuselage.

The subsonic drag of a streamlined, nonlifting body consists solely of skin friction and viscous separation drag and is frequently called the “profile drag.” Profile drag is usually referenced to the maximum cross-sectional area of the body.

Note that the terms “profile drag” and “form drag” are frequently intermixed, although strictly speaking the profile drag is the sum of the form drag and the skin-friction drag. Also note that the term “profile drag” is sometimes used for the zero-lift drag of an airfoil.

Interference drag is the increase in the drag of the various aircraft components due to the change in the airflow caused by other components. For example, the fuselage generally causes an increase in the wing's drag by encouraging airflow separation at the wing root.

Interference drag usually results from an increase in viscous separation, although the skin-friction drag can also be increased if one component causes the airflow over another component to become turbulent or to increase in velocity.

“Wave drag” is the drag caused by the formation of shocks at supersonic and high subsonic speeds. At high subsonic speeds, the shocks form first on the upper surface of the wings because the airflow is accelerated as it passes over the wing.

Drag forces that are a strong function of lift are known as “induced drag” or “drag-due-to-lift.” The induced drag is caused by the circulation about the airfoil that, for a three-dimensional wing, produces vortices in the airflow behind the wing. The energy required to produce these vortices is extracted from the wing as a drag force, and is proportional to the square of the lift.

Another way of looking at induced drag is that the higher-pressure air under the wing escapes around the wingtip to the wing upper surface, reducing the lift and causing the outer part of the wing to fly in an effective downwash. In other words, the wing is always flying uphill! This rotates the lift force vector toward the rear, so that a component of the lift is now in the drag direction.

To counter the pitching moment of the wing, the tail surfaces produce a lift force generally in the downward direction. The induced drag of the tail is called “trim drag.” Trim drag also includes the additional lift required of the wing to counter any download produced by the tail.

When aircraft total drag vs lift is presented, the drag can be calculated with some fixed elevator deflection or it can be calculated using the varying elevator deflections required to trim the aircraft at each lift coefficient. This “trimmed” drag provides the correct data for use in performance calculations.

In supersonic flight there is a component of wave drag that changes as the lift changes. The creation of lift results from changes in the pressure around the aircraft. Wave drag is a pressure drag due to shock formation, and any changes in the pressures around the aircraft will change the location and

strength of the shocks around it resulting in “wave-drag-due-to-lift.” This drag is fairly small and is usually ignored in early conceptual design.

Two-dimensional (2-D) airfoil drag, or profile drag, is a combination of skin-friction drag and viscous separation drag. There is no drag-due-to-lift for the 2-D airfoil because the lift force is perpendicular to the freestream direction. However, the profile drag increases as the angle of attack is increased, leading to some confusion.

This increase in 2-D airfoil drag is due to an increase in viscous separation caused by a greater pressure drop on the upper surface of the airfoil as the angle of attack is increased. This increase in profile drag with increasing angle of attack is not technically caused by the generation of lift, but does vary as the lift is varied.

Most preliminary drag-estimation methods do not actually use the airfoil profile drag data to determine total wing drag. Instead, the drag for an idealized wing with no camber or twist is determined, and then a separate “camber drag” is estimated. Often the camber drag term is included statistically in the drag-due-to-lift calculation even though it is not technically caused by the generation of lift!

Changing the lift on the wing changes the velocities above and below it. This change in local airflow velocity causes a small change in skin-friction drag. Sometimes called a “supervelocity” effect, this is minor and is usually ignored.

12.3 AERODYNAMIC COEFFICIENTS

Lift and drag forces are usually treated as nondimensional coefficients as defined in Eqs. (12.1) and (12.2). The wing reference area, S_{ref} or simply S , is the full trapezoidal area extending to the aircraft centerline. The dynamic pressure of the freestream air is called “ q ,” as defined in Eq. (12.3).

$$L = qSC_L \quad (12.1)$$

$$D = qSC_D \quad (12.2)$$

where

$$q = \frac{1}{2}\rho V^2 \quad (12.3)$$

By definition, the lift force is perpendicular to the flight direction while the drag is parallel to the flight direction. Remember that the 2-D airfoil characteristics are denoted by lowercase subscripts (i.e., C_d) whereas the 3-D wing characteristics are denoted by uppercase subscripts (i.e., C_L).

Drag is normally spoken of as so many “counts” of drag, meaning the four digits to the right of the decimal place. For example, 38 counts of drag mean a drag coefficient of 0.0038.

Figure 12.3 illustrates the drag polar, which is the standard presentation format for aerodynamic data used in performance calculations. The drag polar is simply a plot of the coefficient of lift vs the coefficient of drag.

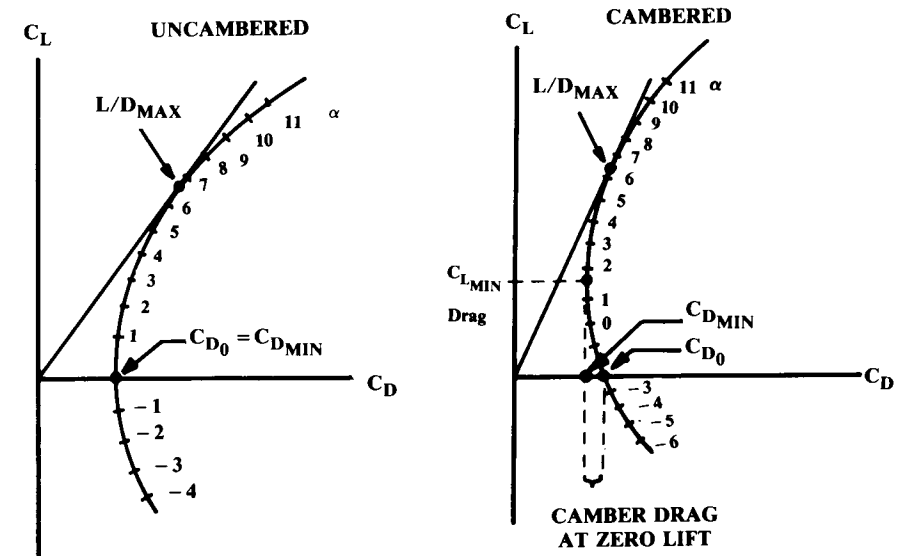


Fig. 12.3 Drag polar.

Note that the angle of attack (α) is indicated here by tic marks along the polar curve. This is not standard practice, but is useful for understanding the relationship between lift, drag, and angle of attack.

$$\text{Uncambered: } C_D = C_{D_0} + KC_L^2 \quad (12.4)$$

$$\text{Cambered: } C_D = C_{D_{\min}} + K(C_L - C_{L_{\min \text{ drag}}})^2 \quad (12.5)$$

For an uncambered wing, the minimum drag (“ C_{D_0} ”) occurs when the lift is zero. The drag polar has an approximately parabolic shape, as defined by Eq. (12.4). The value of “ K ” will be discussed later.

For a cambered wing, the minimum drag (“ $C_{D_{\min}}$ ”) occurs at some positive lift (“ $C_{L_{\min \text{ drag}}}$ ”). The drag polar also has a parabolic shape, but is offset vertically as defined by Eq. (12.5). For wings of moderate camber this offset is usually small, which implies that C_{D_0} approximately equals $C_{D_{\min}}$ and that Eq. (12.4) may be used.

The point at which a line from the origin is just tangent to the drag polar curve is the point of maximum lift-to-drag ratio. Note that this is not the point of minimum drag!

12.4 LIFT

Figure 12.4 shows typical wing lift curves. The uncambered wing has no lift at zero angle of attack, while the cambered wing has a positive lift at zero angle of attack. A negative angle of attack is required to obtain zero lift with a cambered wing.

An old rule-of-thumb is that the negative angle of attack for zero lift in degree equals the airfoil's percent camber (the maximum vertical displacement of the camber line divided by the chord).

Maximum lift is obtained at the "stall" angle of attack, beyond which the lift rapidly reduces. When a wing is stalled, most of the flow over the top has separated.

The slope of the lift curve is essentially linear except near the stall angle, allowing the lift coefficient below stall to be calculated simply as the lift-curve slope times the angle of attack (relative to the zero-lift angle). At the stall, the lift curve has become nonlinear such that the angle for maximum lift is greater than the linear value by an amount shown as $\Delta\alpha$ at $C_{L_{\max}}$ in the figure.

Figure 12.4 also shows the effect of aspect ratio on lift. For an infinite-aspect-ratio wing (the 2-D airfoil case) the theoretical low-speed lift-curve slope is two times π (per radian).

Actual airfoils have lift-curve slopes between about 90 and 100% of the theoretical value. This percentage of the theoretical value is sometimes called the "airfoil efficiency (η)."

Reduction of aspect ratio reduces the lift-curve slope, as shown. At very low aspect ratios, the ability of the air to escape around the wing tips tends to prevent stalling even at very high angles of attack. Also note that the lift curve becomes nonlinear for very low aspect ratios.

Increasing the wing sweep has an effect similar to reducing the aspect ratio. A highly-swept wing has a lift-curve slope much like the aspect-ratio-three curve shown.

The effect of Mach number on the lift-curve slope is shown in Fig. 12.5. The 2-D airfoil lines represent upper boundaries for the no-sweep, infinite aspect-ratio wing. Real wings fall below these curves as shown.

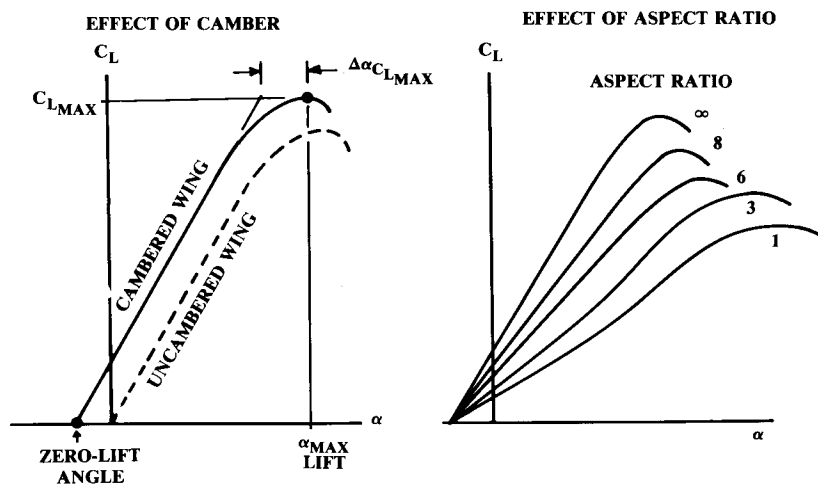


Fig. 12.4 Wing lift curve.

Note that real wings follow a transition curve in the transonic regime between the upward-trending subsonic curve and the downward-trending supersonic curve. Also, note that a fat and unswept wing loses lift in the transonic regime whereas a thinner, swept wing does not.

The lift-curve slope is needed during conceptual design for three reasons. First, it is used to properly set the wing incidence angle. This can be especially important for a transport aircraft, in which the floor must be level during cruise. Also, the wing incidence angle influences the required fuselage angle of attack during takeoff and landing, which affects the aft-fuselage upsweep and/or landing-gear length.

Secondly, the methodology for calculating drag-due-to-lift for high-performance aircraft uses the slope of the lift curve, as will be seen.

The third use of the lift curve slope in conceptual design is for longitudinal-stability analysis, as discussed in Chapter 16.

Subsonic Lift-Curve Slope

Equation (12.6) is a semi-empirical formula from Ref. 36 for the complete wing lift curve slope (per radian). This is accurate up to the drag-divergent Mach number, and reasonably accurate almost to Mach one for a swept wing.

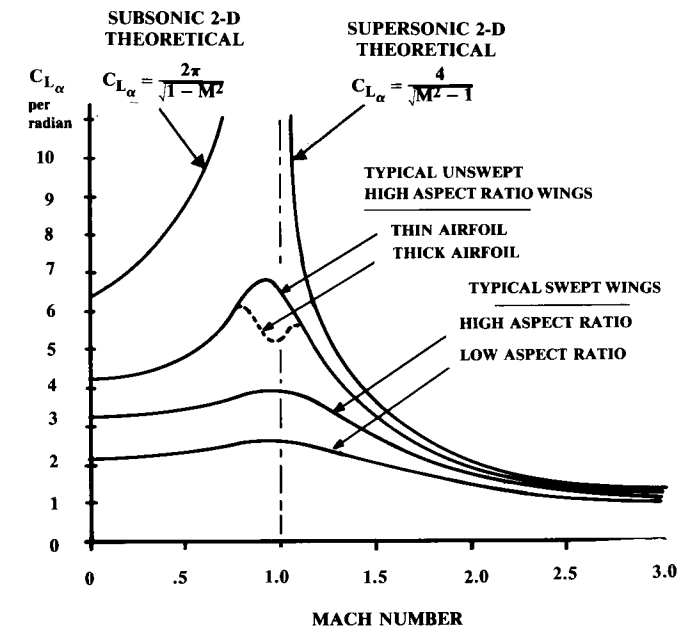


Fig. 12.5 Lift curve slope vs Mach number.

$$C_{L_\alpha} = \frac{2\pi A}{2 + \sqrt{4 + \frac{A^2 \beta^2}{\eta^2} \left(1 + \frac{\tan^2 \Lambda_{\max t}}{\beta^2}\right)}} \left(\frac{S_{\text{exposed}}}{S_{\text{ref}}}\right) (F) \quad (12.6)$$

where

$$\beta^2 = 1 - M^2 \quad (12.7)$$

$$\eta = \frac{C_{l_\alpha}}{2\pi/\beta} \quad (12.8)$$

$\Lambda_{\max t}$ is the sweep of the wing at the chord location where the airfoil is thickest.

If the airfoil lift-curve slope as a function of Mach number is not known, the airfoil efficiency η can be approximated as about 0.95. (In several textbooks this term is dropped by assuming that η equals 1.0 at all Mach numbers.)

“ S_{exposed} ” is the exposed wing planform, i.e., the wing reference area less the part of the wing covered by the fuselage. “ F ” is the fuselage lift factor [Eq. (12.9)] that accounts for the fact that the fuselage of diameter “ d ” creates some lift due to the “spill-over” of lift from the wing.

$$F = 1.07(1 + d/b)^2 \quad (12.9)$$

The wing aspect ratio “ A ” is the geometric aspect ratio of the complete reference planform. The effective aspect ratio will be increased by wing endplates or winglets.

$$\text{Endplate: } A_{\text{effective}} = A(1 + 1.9 h/b) \quad (12.10)$$

where h = endplate height.

$$\text{Winglet: } A_{\text{effective}} \cong 1.2A \quad (12.11)$$

These effective aspect ratios should also be used in the induced drag calculations below. Note that Eq. (12.11) for winglets is a crude approximation based upon limited data for wings of moderate aspect ratio.

The actual increase in effective aspect ratio due to the use of winglets is a function of velocity and lift coefficient and depends upon the selected airfoils and the relative location, geometry, and twist of the wing and winglet. Typically, a wing with higher aspect ratio will obtain less improvement by the use of winglets.

Supersonic Lift-Curve Slope

For a wing in purely supersonic flow, the lift-curve slope is ideally defined by Eq. (12.12), as shown in Fig. 12.5. A wing is considered to be in purely supersonic flow when the leading edge is “supersonic,” i.e., when the Mach cone angle is greater than the leading-edge sweep [see Eq. (12.14)].

$$C_{L_\alpha} = 4/\beta \quad (12.12)$$

where

$$\beta = \sqrt{M^2 - 1} \quad (12.13)$$

when

$$M > 1/\cos \Lambda_{\text{LE}} \quad (12.14)$$

The actual lift-curve slope of a wing in supersonic flight is difficult to predict without use of a sophisticated computer program. The charts in Fig. 12.6 are probably the best approximate method available. They were defined in Ref. 37 and have been used in a number of textbooks.

These charts actually estimate the slope of the “normal force” coefficient (C_n), i.e., the lift curve slope in a direction perpendicular to the surface of the wing. For low angles of attack, this is approximately equal to the lift-curve slope.

To use these charts, the wing aspect ratio, taper ratio, and leading-edge sweep are employed. The six charts each represent data for wings of a different taper ratio. If a chart for the actual taper ratio of a wing is not provided, interpolation must be used.

The term β [Eq. (12.13)] divided by the tangent of the leading-edge sweep is calculated and found on the horizontal axis of the chart. If this ratio is greater than 1.0, it is inverted and the right side of the chart must be used. Then the appropriate line is selected by calculating the wing aspect ratio times the tangent of the leading-edge sweep, and the vertical-axis value is read.

To obtain the approximate slope of the lift curve, this value is then divided by the tangent of the leading-edge sweep, if on the left side of the chart, or by β if on the right side of the chart.

As this value is referenced to the exposed planform of the wing, it must be multiplied by ($S_{\text{exposed}}/S_{\text{ref}}$) as in Eq. (12.6). Also, the value must be multiplied by F from Eq. (12.9) to account for the fuselage lift effect.

Note that these charts give best results only for trapezoidal wings without kinks or strakes. For highly nontrapezoidal planforms, Ref. 37 contains additional estimation procedures. However, these charts are rarely used in industry where computerized “panel methods” are available. These are discussed later.

Transonic Lift-Curve Slope

In the transonic regime (roughly Mach 0.85–1.2 for a swept wing) there are no good initial-estimation methods for slope of the lift curve. It is suggested that the subsonic and supersonic values be plotted vs Mach number, and that a smooth curve be faired between the subsonic and supersonic values similar to the curves shown in Fig. 12.5.

Nonlinear Lift Effects

For a wing of very high sweep or very low aspect ratio (under two or three), the air escaping around the swept leading edge or wing tip will form

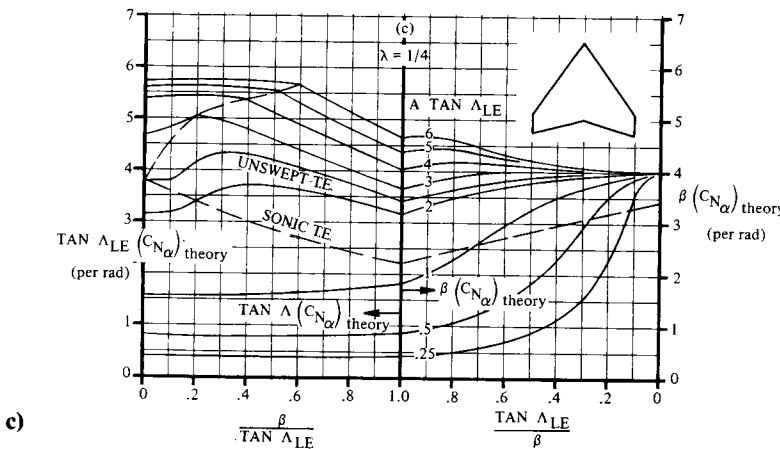
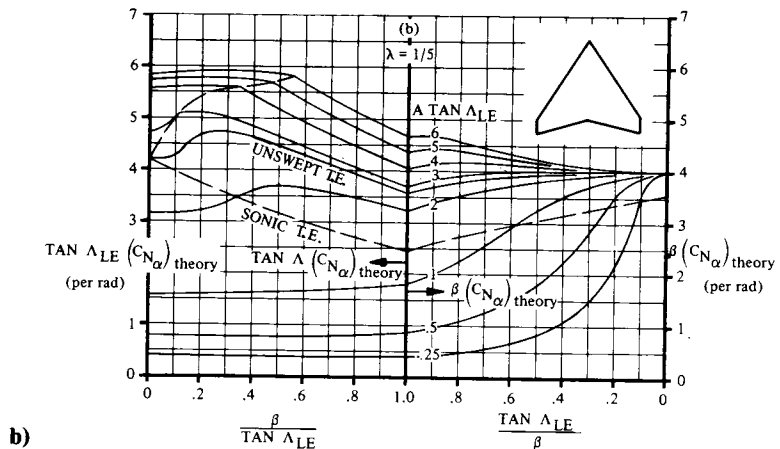
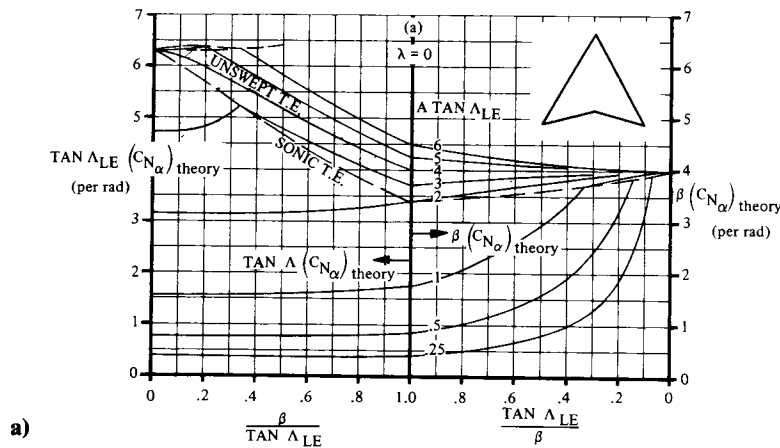


Fig. 12.6 Wing supersonic normal-force-curve slope. (Ref. 37)

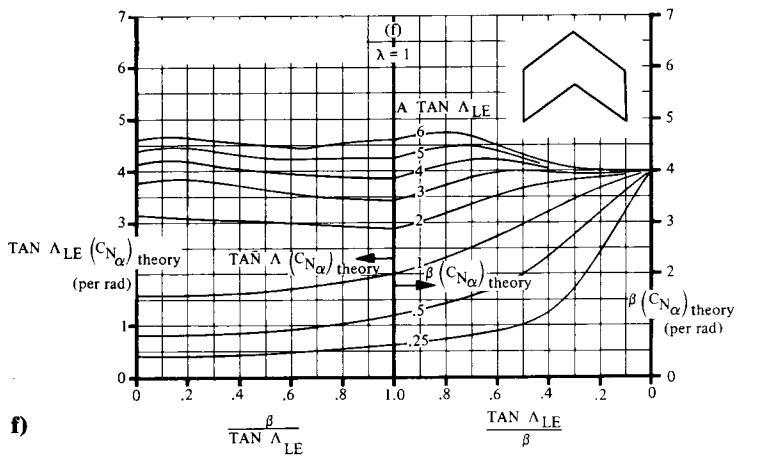
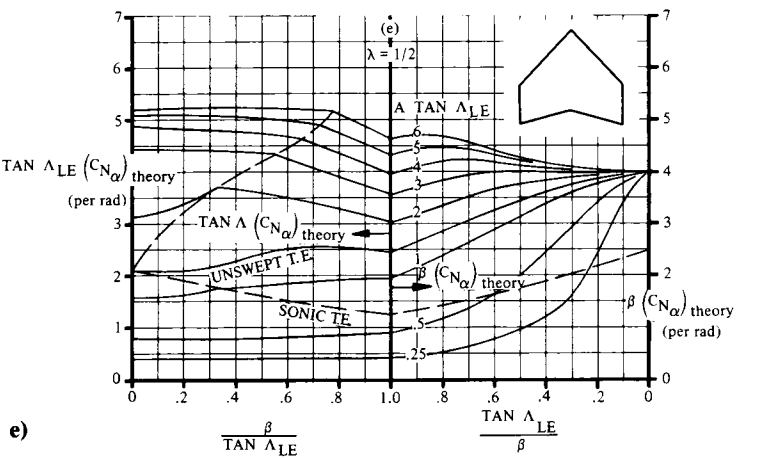
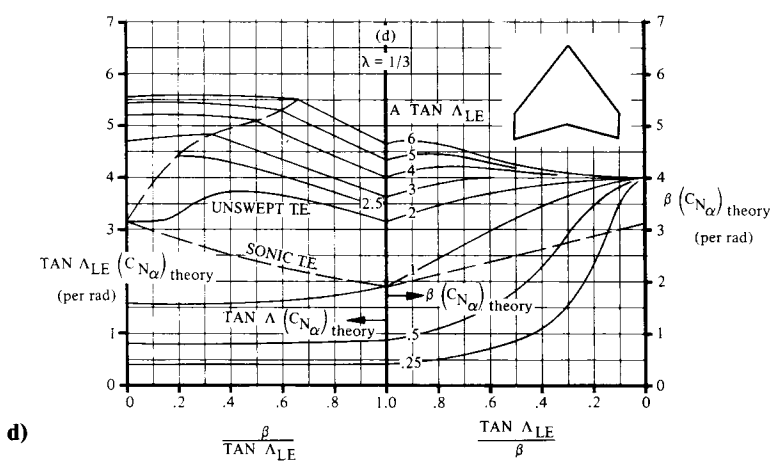


Fig. 12.6 (cont'd.) Wing supersonic normal-force-curve slope. (Ref. 37)

a strong vortex that creates additional lift at a given angle of attack. This additional lift varies approximately by the square of the angle of attack. This nonlinear increase in the slope of the lift curve is difficult to estimate and can conservatively be ignored during early conceptual design.

(However, the increase in maximum lift due to vortex formation is very important. It will be discussed in the next section.)

Maximum Lift (Clean)

The maximum lift coefficient of the wing will usually determine the wing area. This in turn will have a great influence upon the cruise drag. This strongly affects the aircraft takeoff weight to perform the design mission.

Thus, the maximum lift coefficient is critical in determining the aircraft weight; yet the estimation of maximum lift is probably the least reliable of all of the calculations used in aircraft conceptual design. Even refined wind-tunnel tests cannot predict maximum lift with great accuracy. Frequently an aircraft must be modified during flight test to achieve the estimated maximum lift.

For high-aspect-ratio wings with moderate sweep and a large airfoil leading edge radius, the maximum lift depends mostly upon the airfoil characteristics. The maximum lift coefficient of the “clean” wing (i.e., without the use of flaps and other high-lift devices) will usually be about 90% of the airfoil’s maximum lift as determined from the 2-D airfoil data at a similar Reynolds number.

Sweeping the wing reduces the maximum lift, which can be found by multiplying the unswept maximum lift value by the cosine of the quarter-chord sweep [Eq. (12.15)]. This equation is reasonably valid for most subsonic aircraft of moderate sweep.

$$C_{L_{max}} = 0.9 C_{\ell_{max}} \cos \Lambda_{0.25c} \tag{12.15}$$

If a wing has a low aspect ratio or has substantial sweep and a relatively sharp leading edge, the maximum lift will be increased due to the formation of leading-edge vortices. This vortex formation is strongly affected by the shape of the upper surface of the leading edge.

Leading-edge shape could be defined by the airfoil nose radius. However, the nose radius alone doesn’t take into account the effect of airfoil camber on the shape of the upper surface of the airfoil leading edge.

Instead, an arbitrary “leading-edge sharpness parameter” has been defined as the vertical separation between the points on the upper surface, which are 0.15% and 6% of the airfoil chord back from the leading edge (Fig. 12.7). The leading-edge sharpness parameter (or “ Δy ”) as a function of thickness ratio for various airfoils is provided in Table 12.1.

The leading-edge sharpness parameter has been used in Ref. 37 to develop methods for the construction of the lift curve up to the stall, for low- or high-aspect-ratio wings. For high-aspect-ratio wings, Eq. (12.16) is used along with Figs. 12.8 and 12.9. The first term of Eq. (12.16) represents the maximum lift at Mach 0.2, and the second term represents the correction to a higher Mach number.

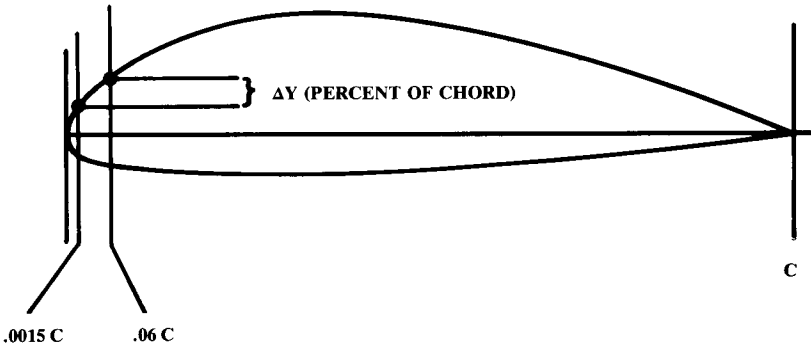


Fig. 12.7 Airfoil leading edge sharpness parameter.

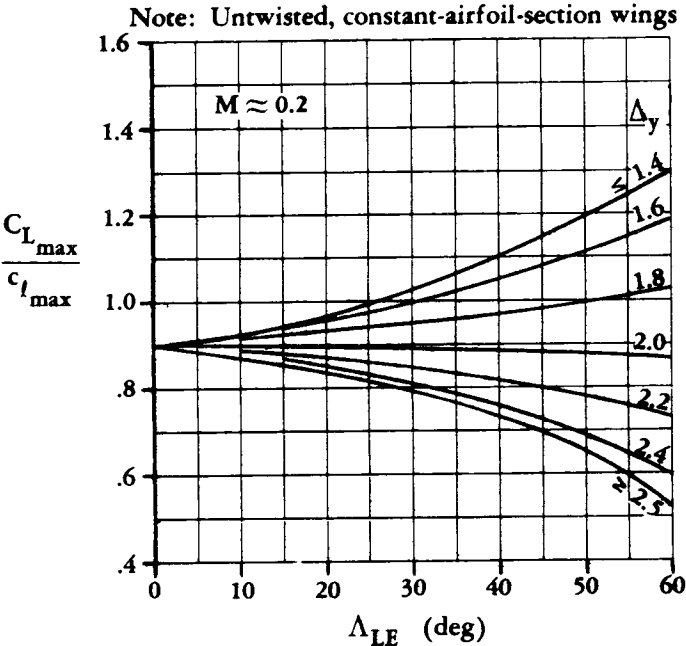


Fig. 12.8 Subsonic maximum lift of high-aspect-ratio wings. (Ref. 37)

Table 12.1 ΔY for common airfoils

Airfoil type	ΔY
NACA 4 digit	26 t/c
NACA 5 digit	26 t/c
NACA 64 series	21.3 t/c
NACA 65 series	19.3 t/c
Biconvex	11.8 t/c

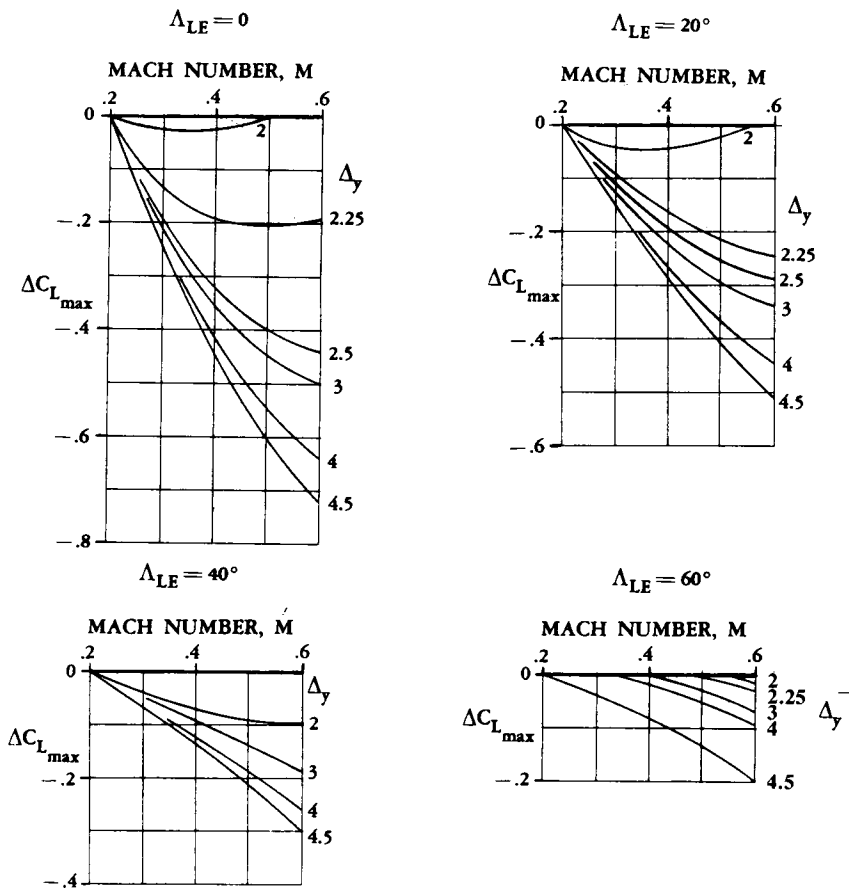


Fig. 12.9 Mach-number correction for subsonic maximum lift of high-aspect-ratio wings. (Ref. 37)

$$\text{High Aspect Ratio: } C_{L_{\max}} = C_{l_{\max}} \left(\frac{C_{L_{\max}}}{C_{l_{\max}}} \right) + \Delta C_{L_{\max}} \quad (12.16)$$

where $C_{l_{\max}}$ is the airfoil maximum lift coefficient at $M = 0.2$.

The angle of attack for maximum lift is defined in Eq. (12.17) with the help of Fig. 12.10. Note that the first and second terms represent the angle of attack if the lift curve slope were linear all the way up to stall. The second term may be approximated by the airfoil zero-lift angle, which is negative for a cambered airfoil. If the wing is twisted, the zero-lift angle is approximately the zero lift angle at the mean chord location. The third term in Eq. (12.17) is a correction for the nonlinear effects of vortex flow.

$$\text{High Aspect Ratio: } \alpha_{C_{L_{\max}}} = \frac{C_{L_{\max}}}{C_{L_{\alpha}}} + \alpha_{0L} + \Delta \alpha_{C_{L_{\max}}} \quad (12.17)$$

A different set of charts is used for a low-aspect-ratio wing, where vortex flow dominates the aerodynamics. For use of these charts, low aspect ratio is defined by Eq. (12.18), which uses the parameter C_1 from Fig. 12.11. Maximum lift of a low-aspect-ratio wing is defined by Eq. (12.19) using Figs. 12.12 and 12.13. The angle of attack at maximum lift is defined by Eq. (12.20) using Figs. 12.14 and 12.15.

$$\text{Low Aspect Ratio if: } A \leq \frac{3}{(C_1 + 1) (\cos \Lambda_{LE})} \quad (12.18)$$

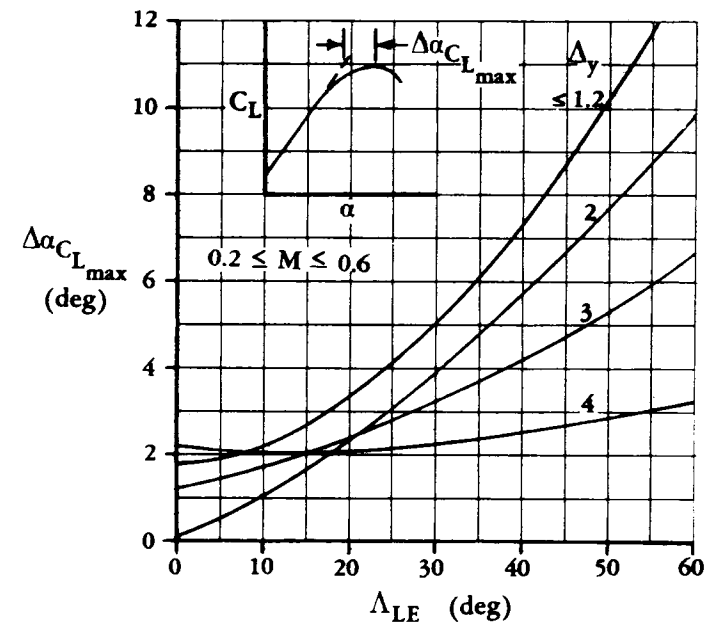


Fig. 12.10 Angle-of-attack increment for subsonic maximum lift of high-aspect-ratio wings. (Ref. 37)

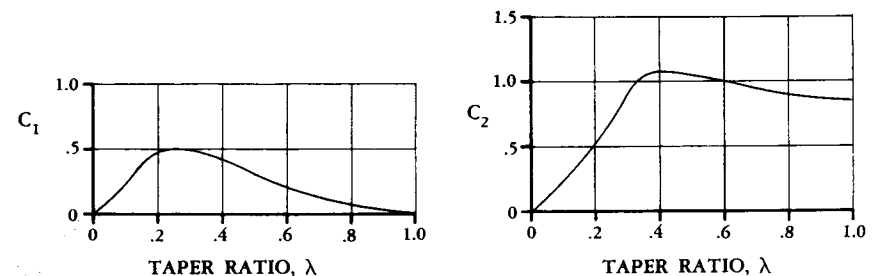


Fig. 12.11 Taper-ratio correction factors for low-aspect-ratio wings. (Ref. 37)

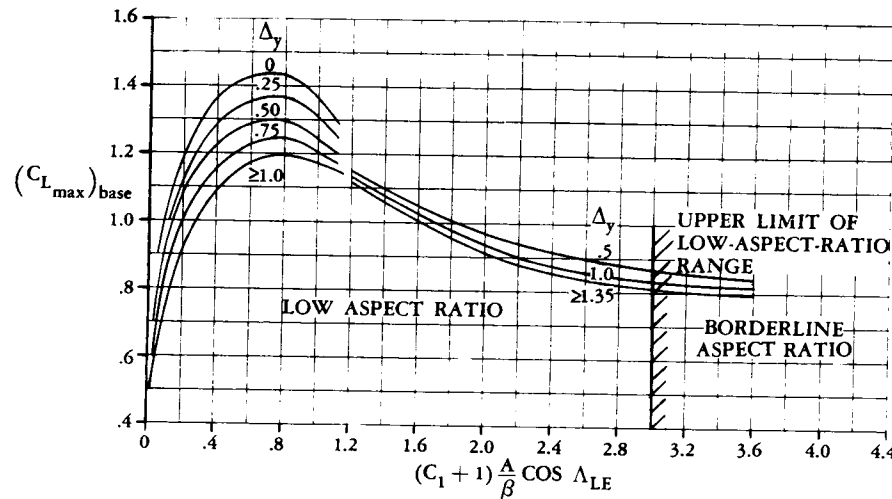


Fig. 12.12 Maximum subsonic lift of low-aspect-ratio wings. (Ref. 37)

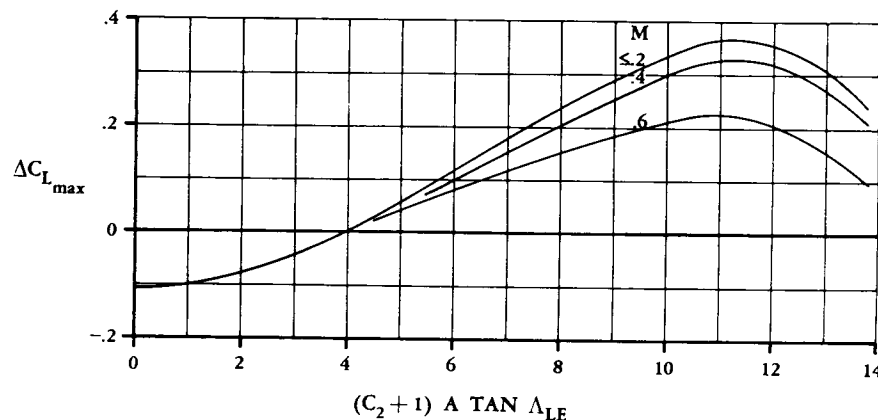


Fig. 12.13 Maximum-lift increment for low-aspect-ratio wings. (Ref. 37)

$$\text{Low Aspect Ratio: } C_{L_{\max}} = (C_{L_{\max}})_{\text{base}} + \Delta C_{L_{\max}} \quad (12.19)$$

$$\alpha_{C_{L_{\max}}} = (\alpha_{C_{L_{\max}}})_{\text{base}} + \Delta \alpha_{C_{L_{\max}}} \quad (12.20)$$

At transonic and supersonic speeds, the maximum lift a wing can achieve is usually limited by structural considerations rather than aerodynamics. Unless the aircraft is flying at a very high altitude, the available maximum lift at Mach 1 is usually enough to break the wings off!

As a conservative assumption, it can be assumed that the maximum lift available at Mach 0.6 will remain constant at higher Mach numbers. Actually, the maximum lift will usually increase through the transonic regime, and then fall again at supersonic speeds.

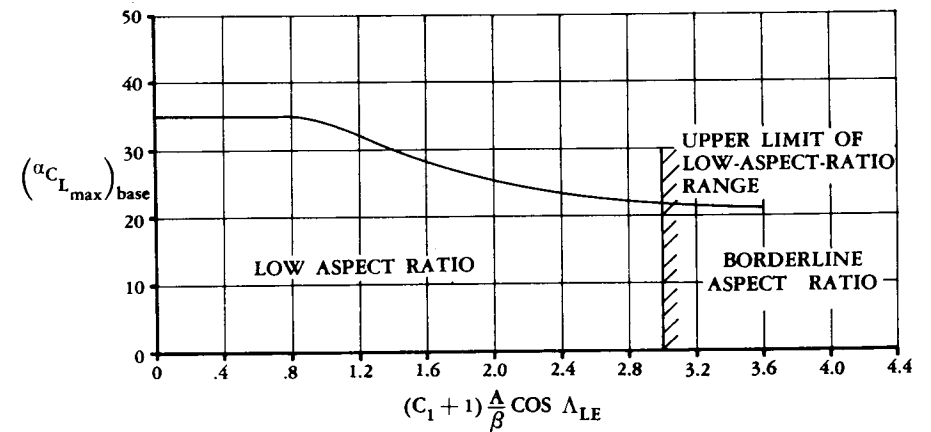


Fig. 12.14 Angle of attack for subsonic maximum lift of low-aspect-ratio wings. (Ref. 37)

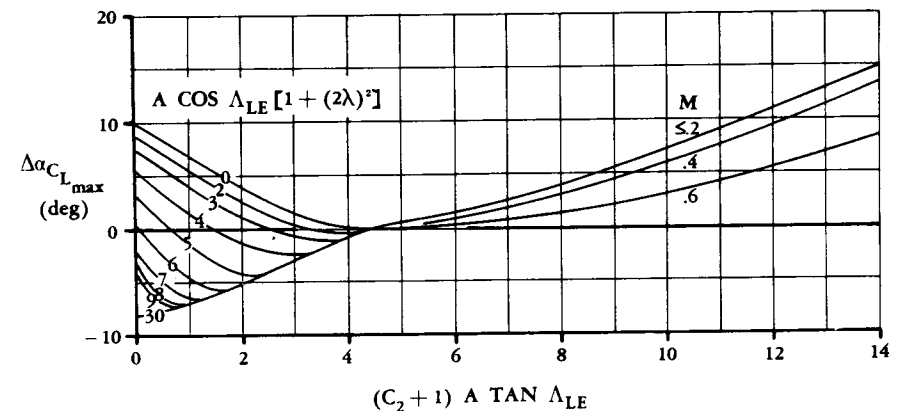


Fig. 12.15 Angle-of-attack increment for subsonic maximum lift of low-aspect-ratio wings. (Ref. 37)

Maximum Lift with High-Lift Devices

There is always a basic incompatibility in aircraft wing design. For cruise efficiency a wing should have little camber and should operate at a high wing-loading. For takeoff and landing a wing should have lots of lift, which means a lot of camber and a low wing-loading.

In the history of aviation almost every imaginable device for varying the wing camber and wing area has been attempted, including a wing with a telescoping outer panel, a fabric membrane which unfurls behind the wing, a device which pivots out from the fuselage forming an extended flap, and even something called a “mutable” wing having variable span, camber, and sweep (Ref. 38).

Figure 12.16 illustrates the commonly used high lift flaps. The plain flap is simply a hinged portion of the airfoil, typically with a flap chord “ C_f ”

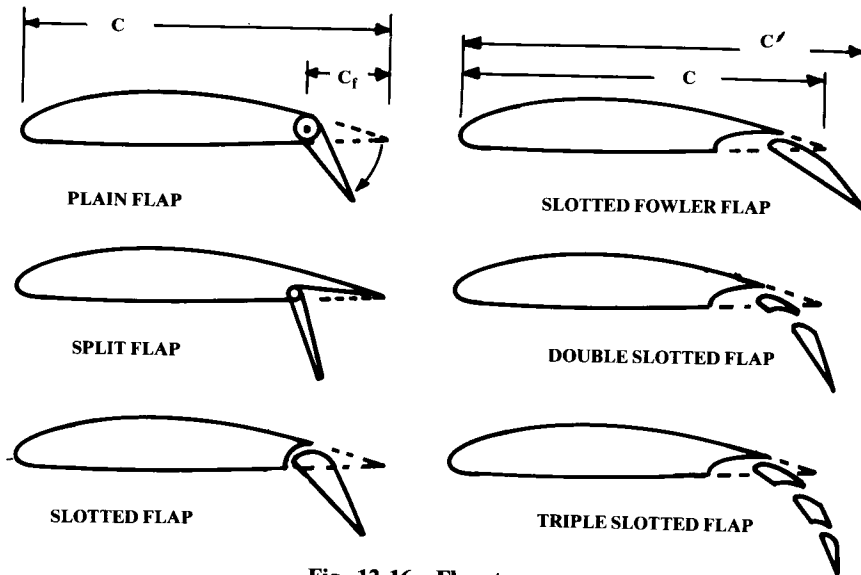


Fig. 12.16 Flap types.

of 30% of the airfoil chord. The plain flap increases lift by increasing camber. For a typical airfoil, the maximum lift occurs with a flap deflection angle of about 40–45 deg. Note that ailerons and other control surfaces are a form of plain flap.

The split flap is like the plain flap except that only the bottom surface of the airfoil is hinged. This produces virtually the same increase in lift as the plain flap. However, the split flap produces more drag and much less change in pitching moment, which may be useful in some designs. Split flaps are rarely used now but were common during World War II.

The slotted flap is a plain flap with a slot between the wing and the flap. This permits high-pressure air from beneath the wing to exit over the top of the flap, which tends to reduce separation. This increases lift and reduces drag.

The “Fowler-type” flap is like a slotted flap, but mechanized to slide rearward as it is deflected. This increases the wing area as well as the camber. Fowler flaps can be mechanized by a simple hinge located below the wing, or by some form of track arrangement contained within it.

To further improve the airflow over the Fowler flap, double- and even triple-slotted flaps are used on some airliners. These increase lift but at a considerable increase in cost and complexity.

Aft flaps do not increase the angle of stall. In fact, they tend to reduce the stall angle by increasing the pressure drop over the top of the airfoil, which promotes flow separation. To increase the stall angle, some form of leading-edge device is required, as shown in Fig. 12.17.

The leading-edge slot is simply a hole which permits high-pressure air from under the wing to blow over the top of the wing, delaying separation and stall. Usually such a slot is fixed, but may have closing doors to reduce drag at high speeds.

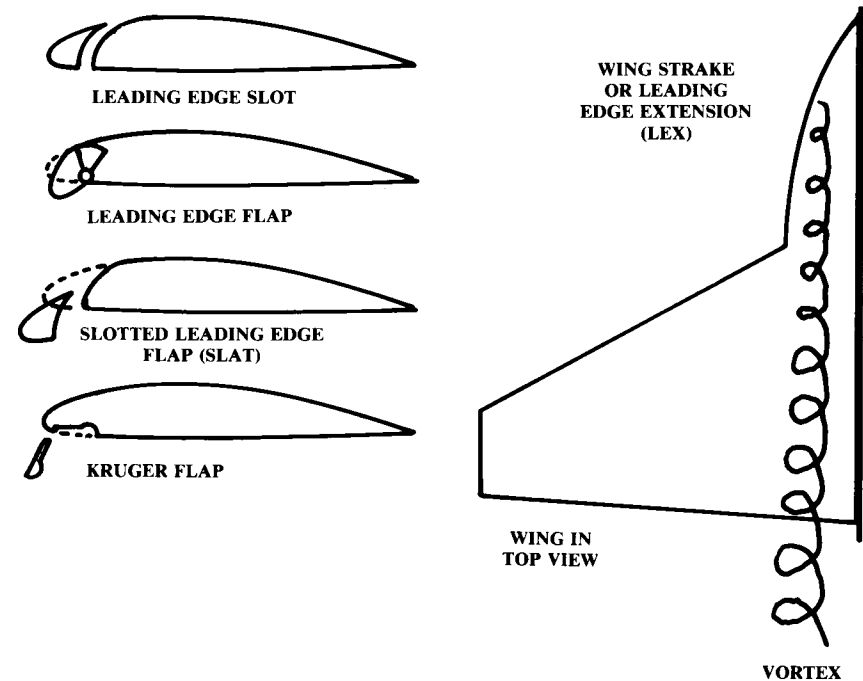


Fig. 12.17 Leading edge devices.

A leading-edge flap is a hinged portion of the leading edge that droops down to increase camber. This has the effect of increasing the curvature on the upper surface. The increase has been shown to be a major factor in determining maximum lift. Leading-edge flaps are usually used for improving the transonic maneuvering performance of high-speed fighters, which need a thin wing for supersonic flight.

A slotted leading-edge flap (“slat”) provides increased camber, a slot, and an increase in wing area. Slat is the most widely used leading-edge device for both low-speed and transonic maneuvering. At transonic speeds, slats are also useful for reducing the buffeting tendency which may limit the usable lift. At Mach 0.9 the use of slats improved the usable lift of the F-4 by over 50%.

The Kruger flap is used mostly by large airliners. It works as an air dam, forcing air up and over the top of the wing. Kruger flaps are lighter in weight than slats but produce higher drag at the lower angles of attack.

The wing strake, or “Leading Edge Extension” (LEX), is similar to the dorsal fin used on vertical tails. Like dorsal fins, the LEX at high angle of attack produces a vortex that delays separation and stall. Unfortunately, a LEX tends to promote pitch-up tendencies and so must be used with care.

Figure 12.18 illustrates the effects these high-lift devices have upon the lift curve of the wing. The nonextending flaps such as the plain, split, or slotted flaps act as an increase in camber, which moves the angle of zero-lift

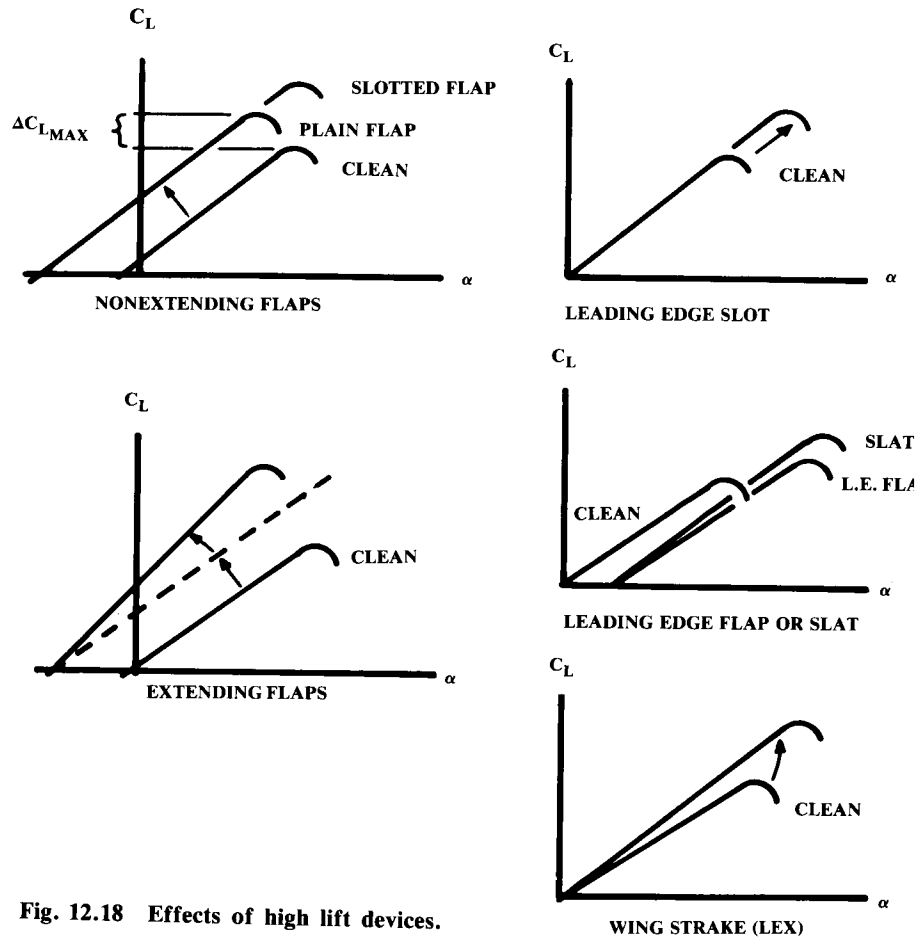


Fig. 12.18 Effects of high lift devices.

to the left and increases the maximum lift. The slope of the lift curve remains unchanged, and the angle of stall is somewhat reduced.

An extending flap such as the Fowler type acts much like the other flaps as far as zero lift angle and stall angle. However, the wing area is increased as the flap deflects, so the wing generates more lift at any given angle of attack compared to the nonextending flap.

Since the lift coefficient is referenced to the original wing area, not the extended wing area, the effective slope of the lift curve for an extending flap is increased by approximately the ratio of the total extended wing area to the original wing area.

Double- and triple-slotted flaps act much like single-slotted Fowler flaps, but the maximum lift is increased.

A leading-edge slot acts only to delay stall. A leading-edge flap or slat delays the stall, but also has the effect of reducing the lift at a given angle of attack (i.e., the lift curve moves to the right). This is because the droop

in the leading edge acts as a reduction in the effective angle of attack as measured from the leading edge to the trailing edge. Note that a leading edge slat, which increases wing area, also increases the slope of the lift curve much as does a Fowler flap.

Leading-edge devices alone do little to improve lift for takeoff and landing, because they are effective only at fairly high angles of attack. However, they are very useful when used in combination with trailing-edge flaps because they prevent premature airflow separation caused by the flaps.

The wing strake, or LEX, delays the stall at high angles of attack (over 20 deg). Also, the area of the LEX provides additional lift, thus increasing the slope of the lift curve. However, the LEX does little to increase lift at the angles of attack seen during takeoff and landing. The LEX does not delay the premature stall associated with trailing-edge flaps.

There are many complex methods for estimating the effects of high-lift devices, some of which are detailed in Ref. 37. For initial design, Eqs. (12.21) and (12.22) provide a reasonable estimate of the increase in maximum lift and the change in the zero-lift angle for flaps and leading-edge devices when deployed at the optimum angle for high lift during landing.

$\Delta C_{L_{max}}$ values should be obtained from test data for the selected airfoil, or may be approximated from Table 12.2. For takeoff flap settings, lift increments of about 60–80% of these values should be used. The change in zero-lift angle for flaps in the 2-D case is approximately -15 deg at the landing setting, and -10 deg at the takeoff setting.

$$\Delta C_{L_{max}} = \Delta C_{L_{max}} \left(\frac{S_{flapped}}{S_{ref}} \right) \cos \Lambda_{H.L.} \quad (12.21)$$

$$\Delta \alpha_{OL} = (\Delta \alpha_{OL})_{airfoil} \left(\frac{S_{flapped}}{S_{ref}} \right) \cos \Lambda_{H.L.} \quad (12.22)$$

Table 12.2 Approximate lift contributions of high-lift devices

High-lift device	$\Delta C_{L_{max}}$
Flaps	
Plain and split	0.9
Slotted	1.3
Fowler	1.3 c'/c
Double slotted	1.6 c'/c
Triple slotted	1.9 c'/c
Leading edge devices	
Fixed slot	0.2
Leading edge flap	0.3
Kruger flap	0.3
Slat	0.4 c'/c

In Eqs. (12.21) and (12.22), “H.L.” refers to the hinge line of the high-lift surface. “ S_{flapped} ” is defined in Fig. 12.19. The lift increment for a leading-edge extension may be crudely estimated as 0.4 at high angles of attack.

Other methods for increasing the lift coefficient involve active flow control using either suction or blowing. Suction uses mechanical air pumps to suck the thickening boundary layer off the wing before it causes separation. This increases the stall angle of attack, and therefore increases maximum lift in a manner similar to leading-edge flaps.

Blowing uses compressor bleed air or compressed air provided by a mechanical air pump to prevent flow separation and increase the freestream-flow turning. Typically, the compressed air is exited through rearward-facing slots over the flaps or leading-edge flaps.

12.5 PARASITE (ZERO-LIFT) DRAG

Equivalent Skin-Friction Method

Two methods for the estimation of the parasite drag (“ C_{D0} ”) are presented below. The first is based upon the fact that a well-designed aircraft in subsonic cruise will have parasite drag that is mostly skin-friction drag plus a small separation pressure drag. The latter is a fairly consistent percentage of the skin-friction drag for different classes of aircraft. This leads to the concept of an “equivalent skin friction coefficient” (C_{fe}), which includes both skin-friction and separation drag.

C_{fe} is multiplied by the aircraft’s wetted area to obtain an initial estimate of parasite drag. This estimate [Eq. (12.23) and Table 12.3] is suitable for initial subsonic analysis and for checking the results of the more detailed method described in the next section.

$$C_{D0} = C_{fe} \frac{S_{\text{wet}}}{S_{\text{ref}}} \quad (12.23)$$

Table 12.3 Equivalent skin friction coefficients

$C_{D0} = C_{fe} \frac{S_{\text{wet}}}{S_{\text{ref}}}$	C_{fe} -subsonic
Bomber and civil transport	0.0030
Military cargo (high upsweep fuselage)	0.0035
Air Force fighter	0.0035
Navy fighter	0.0040
Clean supersonic cruise aircraft	0.0025
Light aircraft – single engine	0.0055
Light aircraft – twin engine	0.0045
Prop seaplane	0.0065
Jet seaplane	0.0040

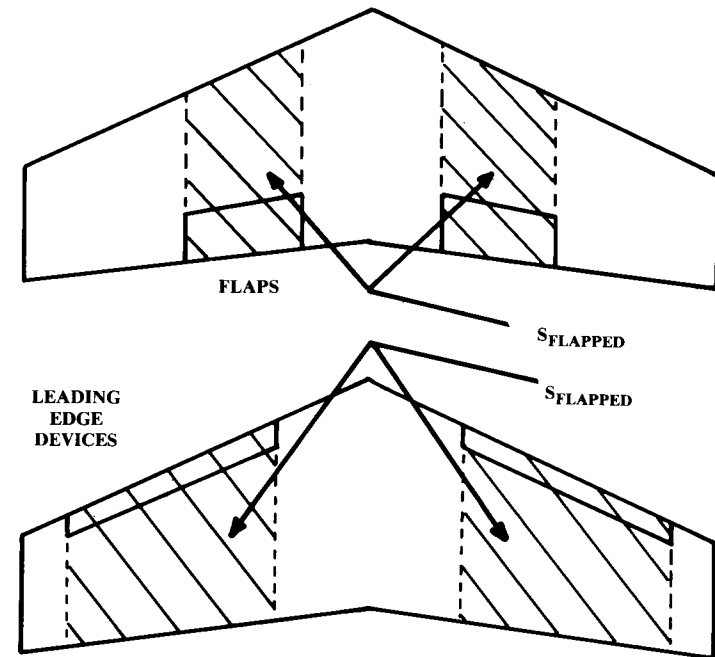


Fig. 12.19 “Flapped” wing area.

Component Buildup Method

The component buildup method estimates the subsonic parasite drag of each component of the aircraft using a calculated flat-plate skin-friction drag coefficient (C_f) and a component “form factor” (FF) that estimates the pressure drag due to viscous separation. Then the interference effects on the component drag are estimated as a factor “ Q ” and the total component drag is determined as the product of the wetted area, C_f , FF , and Q .

(Note that the interference factor Q should not be confused with dynamic pressure q .)

Miscellaneous drags ($C_{D_{\text{misc}}}$) for special features of an aircraft such as flaps, unretracted landing gear, an upswept aft fuselage, and base area are then estimated and added to the total, along with estimated contributions for leakages and protuberances ($C_{D_{\text{L\&P}}}$). Subsonic parasite-drag buildup is shown in Eq. (12.24), where the subscript “c” indicates that those values are different for each component.

$$(C_{D0})_{\text{subsonic}} = \frac{\sum (C_{fe} FF_c Q_c S_{\text{wet}_c})}{S_{\text{ref}}} + C_{D_{\text{misc}}} + C_{D_{\text{L\&P}}} \quad (12.24)$$

For supersonic flight, the skin-friction contribution is simply the flat-plate skin friction coefficient times the wetted area. All supersonic pressure drag contributions (except base drag) are included in the wave-drag term, which is determined from the total aircraft volume distribution.

For transonic flight, a graphical interpolation between subsonic and supersonic values is used. Supersonic and transonic drag calculations are discussed later.

Flat-Plate Skin Friction Coefficient

The flat-plate skin friction coefficient C_f depends upon the Reynolds number, Mach number, and skin roughness. The most important factor affecting skin-friction drag is the extent to which the aircraft has laminar flow over its surfaces.

At a local Reynolds number of one million, a surface with turbulent flow will have a friction drag coefficient as much as three times the drag coefficient of a surface with laminar flow. Laminar flow may be maintained if the local Reynolds number is below roughly half a million, and only if the skin is very smooth (molded composite or polished aluminum without rivets).

Most current aircraft have turbulent flow over virtually the entire wetted surface, although some laminar flow may be seen towards the front of the wings and tails. A typical current aircraft may have laminar flow over perhaps 10–20% of the wings and tails, and virtually no laminar flow over the fuselage.

A carefully designed modern composite aircraft such as the Piaggio GP180 can have laminar flow over as much as 50% of the wings and tails, and about 20–35% of the fuselage.

For the portion of the aircraft that has laminar flow, the flat-plate skin friction coefficient is expressed by Eq. (12.25). Note that laminar flow is unlikely at transonic or supersonic speeds, unless great attention is paid to shaping and surface smoothness.

$$\text{Laminar: } C_f = 1.328/\sqrt{R} \quad (12.25)$$

where Reynolds number is:

$$R = \rho V \ell / \mu \quad (12.26)$$

The “ ℓ ” in Eq. (12.26) is the characteristic length. For a fuselage, ℓ is the total length. For a wing or tail, ℓ is the mean aerodynamic chord length.

For turbulent flow, which in most cases covers the whole aircraft, the flat-plate skin friction coefficient is determined by Eq. (12.27). Note that the second term in the denominator, the Mach number correction, goes to 1.0 for low-subsonic flight.

$$\text{Turbulent: } C_f = \frac{0.455}{(\log_{10} R)^{2.58} (1 + 0.144 M^2)^{0.65}} \quad (12.27)$$

If the surface is relatively rough, the friction coefficient will be higher than indicated by this equation. This is accounted for by the use of a “cut-off Reynolds number,” which is determined from Eq. (12.28) or (12.29) using the characteristic length ℓ (feet) and a skin-roughness value “ k ” based upon Table 12.4. The lower of the actual Reynolds number and the cut-off Reynolds number should be used in Eq. (12.27).

$$\text{Subsonic: } R_{\text{cutoff}} = 38.21(\ell/k)^{1.053} \quad (12.28)$$

$$\text{Transonic or Supersonic: } R_{\text{cutoff}} = 44.62(\ell/k)^{1.053} M^{1.16} \quad (12.29)$$

Once laminar and turbulent flat-plate skin friction coefficients have been calculated, an “average” coefficient can be calculated as the weighted average of the two. This requires estimation of the percentage of laminar flow which can be attained. This estimation is a judgment call based on past experience as discussed above, and one must review the current literature to determine how much laminar flow can be attained with current state of the art.

Component Form Factors

Form factors for subsonic-drag estimation are presented in Eqs. (12.30–12.32). These are considered valid up to the drag-divergent Mach number. In Eq. (12.30), the term “ $(x/c)_m$ ” is the chordwise location of the airfoil maximum thickness point. For most low-speed airfoils, this is at about 0.3 of the chord. For high-speed airfoils this is at about 0.5 of the chord. Λ_m refers to the sweep of the maximum-thickness line.

Wing, Tail, Strut, and Pylon:

$$FF = \left[1 + \frac{0.6}{(x/c)_m} \left(\frac{t}{c} \right) + 100 \left(\frac{t}{c} \right)^4 \right] \left[1.34 M^{0.18} (\cos \Lambda_m)^{0.28} \right] \quad (12.30)$$

Fuselage and Smooth Canopy:

$$FF = \left(1 + \frac{60}{f^3} + \frac{f}{400} \right) \quad (12.31)$$

Nacelle and Smooth External Store:

$$FF = 1 + (0.35/f) \quad (12.32)$$

where

$$f = \frac{\ell}{d} = \frac{\ell}{\sqrt{(4/\pi) A_{\text{max}}}} \quad (12.33)$$

A tail surface with a hinged rudder or elevator will have a form factor about 10% higher than predicted by Eq. (12.30) due to the extra drag of the gap between the tail surface and its control surface.

Table 12.4 Skin roughness value (k)

Surface	k (ft)
Camouflage paint on aluminum	3.33×10^{-5}
Smooth paint	2.08×10^{-5}
Production sheet metal	1.33×10^{-5}
Polished sheet metal	0.50×10^{-5}
Smooth molded composite	0.17×10^{-5}

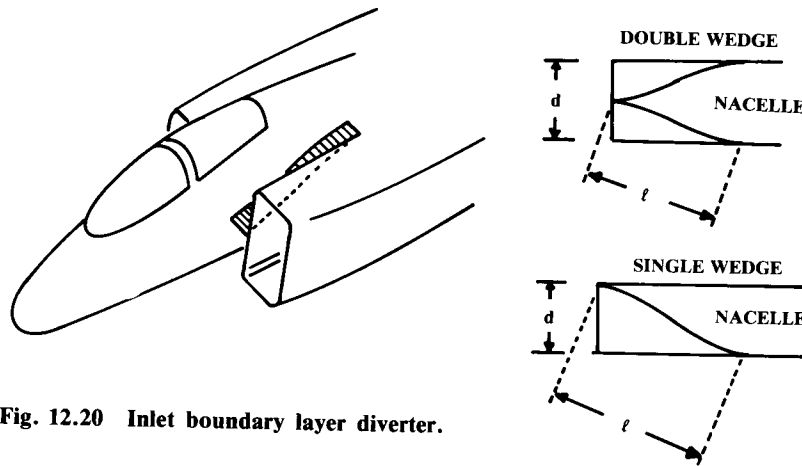


Fig. 12.20 Inlet boundary layer diverter.

Equation (12.31) is mainly used for estimation of the fuselage form factor, but can also be used for a blister or fairing such as a pod used for landing-gear stowage.

For a fuselage with a steep aft-fuselage closure angle in front of a pusher propeller, the separation drag will be lower than predicted using this form-factor equation.

A square-sided fuselage has a form factor about 40% higher than the value estimated with Eq. (12.31) due to additional separation caused by the corners. This can be somewhat reduced by rounding the corners. A flying-boat hull has a form factor about 50% higher, and a float has a form factor about three times the estimated value.

Equation (12.31) will predict the form factor for a smooth, one-piece fighter canopy such as seen on the F-16. For a typical two-piece canopy with a fixed but streamlined windscreen (i.e., F-15), the form factor calculated with Eq. (12.31) should be increased by about 40%. A canopy with a flat-sided windscreen has a form factor about three times the value estimated with Eq. (12.31).

The external boundary-layer diverter for an inlet mounted on the fuselage can have a large drag contribution. Equations (12.34) and (12.35) estimate the form factors to use for a double-wedge and single-wedge diverter, where the Reynolds number is determined using ℓ and the wetted area is defined as shown in Fig. 12.20. Remember to double the drag if there are two inlets.

$$\text{Double Wedge: } FF = 1 + (d/\ell) \quad (12.34)$$

$$\text{Single Wedge: } FF = 1 + (2d/\ell) \quad (12.35)$$

Component Interference Factors

Parasite drag is increased due to the mutual interference between components. For a nacelle or external store mounted directly on the fuselage or wing, the interference factor Q is about 1.5. If the nacelle or store is mounted less than about one diameter away, the Q factor is about 1.3. If it

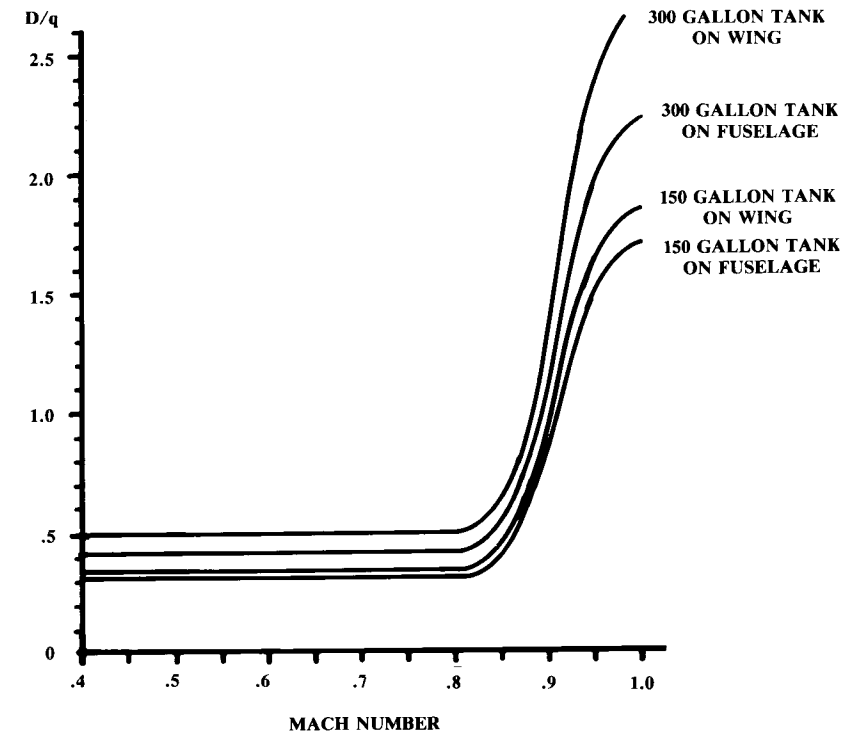


Fig. 12.21 External stores drag-fuel tanks.

is mounted much beyond one diameter, the Q factor approaches 1.0. Wing tip-mounted missiles have a Q factor of about 1.25.

For a high-wing, a mid-wing, or a well-filletted low wing, the interference will be negligible so the Q factor will be about 1.0. An unfilletted low wing can have a Q factor from about 1.1–1.4.

The fuselage has a negligible interference factor ($Q = 1.0$) in most cases. Also, $Q = 1.0$ for a boundary-layer diverter. For tail surfaces, interference ranges from about three percent ($Q = 1.03$) for a clean V-tail to about eight percent for an H-tail. For a conventional tail, four to five percent may be assumed (Ref. 8).

Component parasite drags can now be determined using Eq. (12.24) and the skin-friction coefficients, form factors, and interference factors.

Miscellaneous Drags

The drag of miscellaneous items can be determined separately using a variety of empirical graphs and equations, and then adding the results to the parasite drags determined above.

While the drag of smooth external stores can be estimated using Eq. (12.31), the majority of external stores are in fact not very smooth. Figures 12.21 and 12.22 provide drag estimates for external fuel tanks and weapons, presented as drag divided by dynamic pressure (D -over- q or D/q).

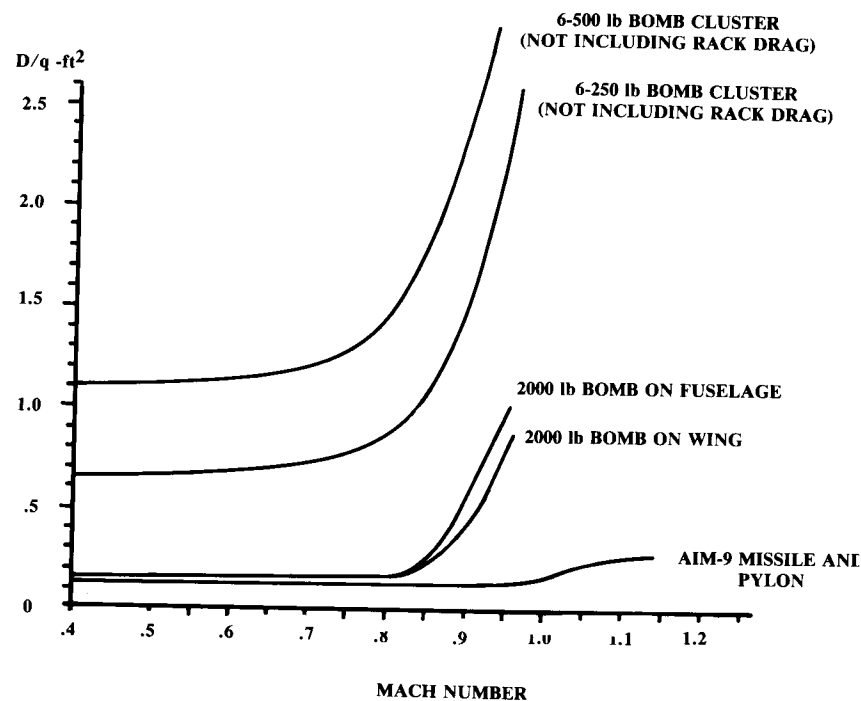


Fig. 12.22 Bomb and missile drag.

D/q has units of square feet, and so is sometimes called the “drag area.” D/q divided by the wing reference area yields the miscellaneous parasite drag coefficient. Note that pylon and bomb-rack drag as estimated using Fig. 12.23 must be added to the store drag.

Most transport and cargo aircraft have a pronounced upsweep to the aft fuselage (Fig. 12.24). This increases the drag beyond the value calculated using Eq. (12.31). This extra drag is a complicated function of the fuselage cross-sectional shape and the aircraft angle of attack, but can be approximated using Eq. (12.36) where “ u ” is the upsweep angle (radians) of the fuselage centerline and $A_{\max}$ is the maximum cross-sectional area of the fuselage.

$$D/q_{\text{upsweep}} = 3.83u^{2.5}A_{\max} \quad (12.36)$$

The landing-gear drag is best estimated by comparison to test data for a similar gear arrangement. Such data for a variety of aircraft is available in Refs. 7, 8, 28, and others. If such data is not available, the gear drag can be estimated as the summation of the drags of the wheels, struts, and other gear components using the data in Table 12.5 (largely from Ref. 8).

These values times the frontal area of the indicated component yield D/q values, which must be divided by the wing reference area to obtain parasite-drag coefficients. To account for mutual interference it is suggested that the sum of the gear component drags be multiplied by 1.2. Also, the total gear drag should be increased by about 7% for a retractable landing gear in which the gear wells are left open when the gear is down.

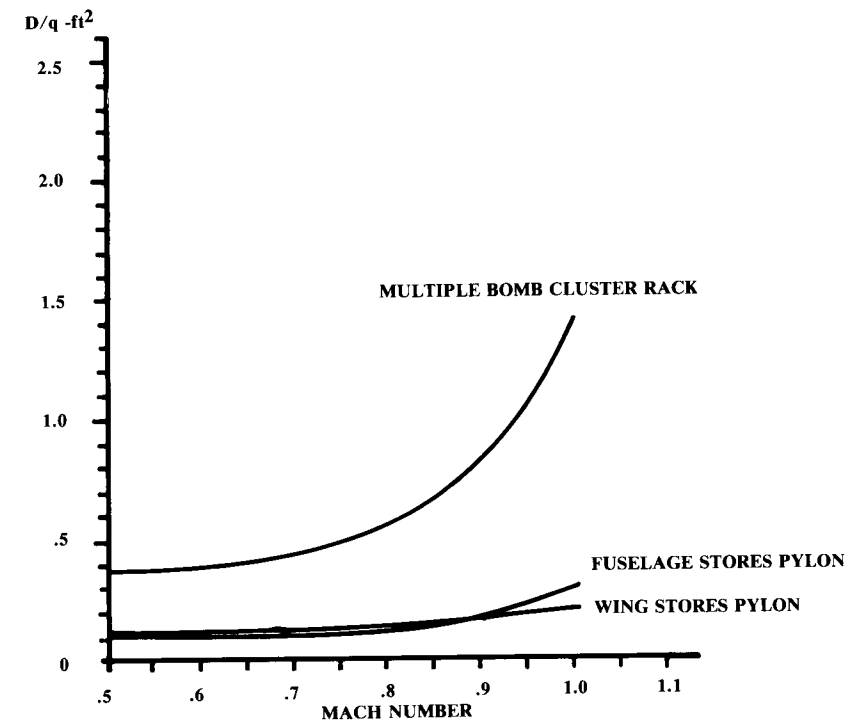


Fig. 12.23 Pylon and bomb rack drag.



Fig. 12.24 Fuselage upsweep.

Table 12.5 Landing gear component drags

	D/q Frontal area (Ft ²)
Regular wheel and tire	0.25
Second wheel and tire in tandem	0.15
Streamlined wheel and tire	0.18
Wheel and tire with fairing	0.13
Streamline strut ($1/6 < t/c < 1/3$)	0.05
Round strut or wire	0.30
Flat spring gear leg	1.40
Fork, bogey, irregular fitting	1.0–1.4

Note that landing-gear drag is actually a function of lift. The more lift the aircraft wing is producing, the greater the velocity of the airflow over the top of the wing and, conversely, the lesser the airflow velocity underneath the wing where the gear is located. Hence, at higher lift coefficients the gear drag is reduced. This can be ignored for initial analysis.

Strut, wire, and fitting data in Table 12.4 may also be used to estimate the extra drag for a braced wing or biplane. The optimal thickness ratio considering both aerodynamic and structural efficiency is about 0.19 for a strut in tension and about 0.23 for a strut in compression.

Flaps affect both the parasite and induced drag. The induced effect is due to the change in the lift distribution, but is relatively small and can be ignored for initial analysis.

The flap contribution to parasite drag is caused by the separated flow above the flap, and can be estimated using Eq. (12.37) for most types of flap. Note that this is referenced to wing area. Typically the flap deflection is about 60–70 deg for landing and about 20–40 deg for takeoff. Light aircraft usually take off with no flaps.

$$\Delta C_{D_{0\text{flap}}} = 0.0023 \frac{\text{flap span}}{\text{wing span}} \delta_{\text{flap}} \quad (12.37)$$

where δ_{flap} is in deg. Note that this is a very rough estimate!

Many aircraft have some form of speed brake. Typically these are plates which extend from the fuselage or wing. Fuselage-mounted speed brakes have a D/q of about 1.0 times the speed-brake frontal area, while wing-mounted speed brakes have a D/q of about 1.6 times their frontal area if mounted at about the 60% of chord location.

Speed brakes mounted on top of the wing will also disturb the airflow and spoil the lift, and so are called “spoilers.” These further reduce landing distance by transferring more of the aircraft’s weight to the landing-gear which increases the braking action.

Base area produces a drag according to Eqs. (12.38) and (12.39) (Ref. 40). “ A_{base} ” includes any aft-facing flat surfaces as well as the projected aft-facing area for any portions of the aft fuselage that experience highly-separated airflow. Roughly speaking, this should be expected any place where the aft fuselage angle to the freestream exceeds about 20 deg. As previously mentioned, a pusher propeller may prevent aft-fuselage separation despite an aft fuselage angle of 30 deg or more.

$$\text{Subsonic: } (D/q)_{\text{base}} = [0.139 + 0.419(M - 0.161)^2] A_{\text{base}} \quad (12.38)$$

$$\text{Supersonic: } (D/q)_{\text{base}} = [0.064 + 0.042(M - 3.84)^2] A_{\text{base}} \quad (12.39)$$

Fighter-type canopies have been discussed above. For transport and light-aircraft windshields that smoothly fair into the fuselage, an additional D/q of about 0.07 times the windshield frontal area is suggested. A sharp-edged, poorly-faired windshield has an additional D/q of about 0.15 times its frontal area.

An open cockpit has a D/q of about 0.50 times the windshield frontal area. For an aircraft with an unenclosed cockpit, such as a hang-glider or ultralight, a seated person has a D/q of about 6 ft². This reduces to a D/q of 1.2 ft² in the prone position.

An arresting hook for carrier operation adds a D/q of about 0.15 ft². The smaller emergency arresting hook for Air Force aircraft adds a D/q of about 0.10 ft². Machine-gun ports add a D/q of about 0.02 ft² per gun. A cannon port such as for the M61 adds a D/q of about 0.2 ft².

Leakage and Protuberance Drag

Leaks and protuberances add drag that is difficult to predict by any method. Leakage drag is due to the tendency of an aircraft to “inhale” through holes and gaps in high-pressure zones, and “exhale” into the low-pressure zones. The momentum loss of the air “inhaled” contributes directly to drag, and the air “exhaled” tends to produce additional airflow separation.

Protuberances include antennas, lights, and such manufacturing defects as protruding rivets and rough or misaligned skin panels. Typically these drag increments are estimated as a percent of the total parasite drag.

For a normal production aircraft, leaks and protuberance drags can be estimated as about 2–5% of the parasite drag for jet transports or bombers, 5–10% for propeller aircraft, and 10–15% for current-design fighters (5–10% for new-design fighters). If special care is taken during design and manufacturing, these drag increments can be reduced to near zero but at a considerable expense.

An aircraft with variable-sweep wings will have an additional protuberance drag of about 3% due to the gaps and steps of the wing pivot area.

Stopped-Propeller and Windmilling Engine Drags

The specifications for civilian and military aircraft require takeoff and climb capabilities following an engine failure. Not only does this reduce the available thrust, but the drag of the stopped propeller or windmilling engine must be considered.

Data on the drag of a stopped or windmilling propeller are normally obtained from the manufacturer. For a jet engine, detailed knowledge of the characteristics of the engine, inlet, and nozzle are required to estimate the drag from a stopped or windmilling engine. In the absence of such data, the following rough approximations can be used.

For a stopped propeller, Ref. 8 indicates that the subsonic drag coefficient will be about 0.1 based upon the total blade area if the propeller is feathered (turned so that the blades align with the airflow). If the propeller has fixed pitch and cannot be feathered, the drag coefficient is about 0.8.

To determine the total blade area it is necessary to know or to estimate the propeller “solidity” (σ), the ratio between the total blade area and the propeller disk area. This can be shown to equal the number of blades divided by the blade aspect ratio and π .

For a typical blade aspect ratio of 8, the solidity will be 0.04 times the number of blades. A small piston-prop engine will generally use a two-

bladed propeller. A fast piston-prop or a small turboprop will use a three-bladed propeller, while a large turboprop may use a four-bladed propeller. Drag of a feathered propeller can be roughly estimated by Eq. (12.40). For an unfeathered, stopped propeller, the 0.1 term is replaced by 0.8.

$$(D/q)_{\text{feathered prop}} = 0.1 \sigma A_{\text{propeller disk}} \quad (12.40)$$

For jet engines, Ref. 9 indicates that the subsonic drag coefficient of a windmilling turbojet engine will be about 0.3, referenced to the flow area at the engine's front face. Thus, the drag of a windmilling turbojet will be approximately:

$$(D/q)_{\text{windmilling jet}} = 0.3 A_{\text{engine front face}} \quad (12.41)$$

Supersonic Parasite Drag

The supersonic parasite drag is calculated in a similar fashion to the subsonic drag, with two exceptions. First, the supersonic skin-friction drag does not include adjustments for form factors or interference effects (i.e., $FF = Q = 1.0$). Second, a new term, wave drag, is added. This accounts for the pressure drag due to shock formation. Supersonic parasite-drag buildup is defined in Eq. (12.42).

$$C_{D0\text{supersonic}} = \frac{\Sigma(C_{fc} S_{wet_c})}{S_{\text{ref}}} + C_{D\text{misc}} + C_{D_{L\&P}} + C_{D\text{wave}} \quad (12.42)$$

The supersonic turbulent skin friction coefficient was previously presented in Eq. (12.27), using the cutoff Reynolds number from Eq. (12.29).

Miscellaneous drag calculations for supersonic flight have been presented above, where appropriate. Many of the items that produce miscellaneous drag will not appear on a supersonic aircraft (floats, open cockpits, etc.).

The drag due to leaks and protuberances in supersonic flight follows about the same percentages presented above, applied to the skin-friction drag only.

The wave drag in supersonic flight will often be greater than all the other drag put together. Wave drag is pressure drag due to shocks, and is a direct result of the way in which the aircraft's volume is distributed.

An ideal volume distribution is produced by the Sears-Haack body (Ref. 16), which was shown in Fig. 8.2. A Sears-Haack body, as defined by Eq. (12.43), has a wave drag as in Eq. (12.45). This is the minimum possible wave drag for any closed-end body of the same length and total volume.

$$\frac{r}{r_{\text{max}}} = \left[1 - \left(\frac{x}{\ell/2} \right)^2 \right]^{0.75} \quad (12.43)$$

where

r = the cross-section radius
 ℓ = the longitudinal dimension

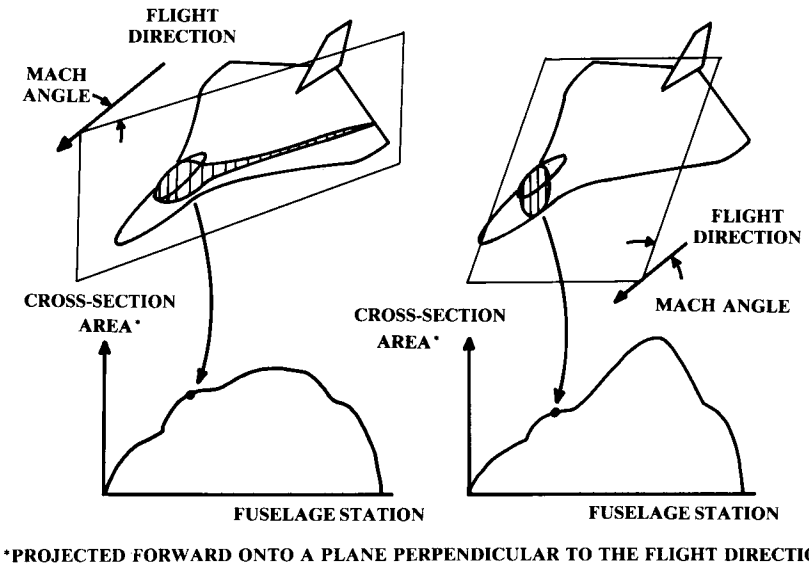


Fig. 12.25 Mach-plane cut volume distribution (two roll angles).

and

$$-\ell/2 \leq x \leq \ell/2 \quad (12.44)$$

$$(D/q)_{\text{wave}} = \frac{9\pi}{2} \left(\frac{A_{\text{max}}}{\ell} \right)^2 \quad (12.45)$$

where A_{max} is the maximum cross-sectional area.

The linear area-rule theory says that the theoretical wave drag of an aircraft at Mach 1.0 is identical to the wave drag of a body of revolution with the same volume-distribution plot. In other words, the actual cross-sectional shape at a given longitudinal location has no effect on wave drag at Mach 1.0. All that matters is the cross-sectional area at each longitudinal location and the way that the cross-sectional area varies longitudinally.

This leads to the area-rule principle for minimizing wave drag. Wave drag at Mach 1.0 is minimized when the aircraft has a volume distribution identical to that of a Sears-Haack body. Drag is reduced when the volume distribution is changed to more resemble the Sears-Haack's, which has a minimal amount of longitudinal curvature.

As was discussed in Chapter 8, the wave drag at Mach 1.0 is directly related to the second derivative (i.e., curvature) of the longitudinal volume distribution. To minimize wave drag the designer should try to arrange the configuration so that the volume distribution is smooth and bell-shaped.

For a typical aircraft, the wing tends to put a "bump" in the volume distribution. This bump can be reduced by pinching in the fuselage at the wing location, creating the characteristic "coke-bottle" area-ruled fuselage.

No realistic aircraft will have a volume distribution identical to that of a Sears-Haack body. However, a well-designed supersonic aircraft will have

a theoretical wave drag at Mach 1.0 that is less than twice the Sears-Haack value. Typical ratios of actual wave drag to the optimum Sears-Haack value will be used below as a first-order wave drag estimation method.

At Mach 1.0, shocks form at an angle of 90 deg to the freestream direction. At Mach numbers higher than 1.0, the shocks may form at an angle less than 90 deg. The "Mach angle" is the smallest angle at which a shock may form, representing a "zero-strength" shock. Mach angle is defined as arcsine $(1/M)$.

At Mach 1.0, the wave drag is based upon the aircraft's cross-sectional areas found by the intersection of the aircraft and an infinite plane set at an angle perpendicular (90 deg) to the freestream direction. At speeds higher than Mach 1.0, the wave drag still depends upon the volume distribution as before, but with one major exception.

At higher Mach numbers the volume distribution is based upon aircraft cross sections that are determined by intersecting the aircraft with "Mach planes," set at the appropriate Mach angle to the freestream direction.

A Mach plane may be rolled about the freestream direction to any roll angle. Figure 12.25 shows two roll angles. Note that the different Mach-plane roll angles produce entirely different volume-distribution plots. In the left illustration, the Mach-plane cut includes the fuselage and canopy plus a slice of the left wing. In the right illustration only the fuselage and canopy are cut, producing a much smaller cross-sectional area at that location.

For each Mach-plane roll angle, a volume-distribution plot can be prepared by taking Mach-plane cuts at a number of longitudinal locations. According to linear wave drag theory (Ref. 41) the supersonic wave drag at Mach numbers greater than 1.0 is determined by averaging the wave drags of the Mach-plane-cut volume distributions for different roll angles.

This is the basis of the widely-used Harris Wave Drag code (Ref. 42). A simplified computer code suitable for university use is presented in Ref. 43.

The use of canted Mach-plane cuts to determine the volume distribution at Mach numbers greater than 1.0 requires a different approach to area-ruling. Pinching the fuselage at the wing location may smooth out the volume distribution for one Mach-plane roll angle, but may make the volume distribution even less smooth at another Mach-plane roll angle.

At higher Mach numbers it is very difficult to minimize total wave drag by "eyeball" area-ruling. Instead it is more profitable to smooth the entire configuration through wing-body blending, as seen on the B-1B and in the design concept of Fig. 7.3.

For preliminary wave drag analysis at $M \geq 1.2$, without use of a computer, a correlation to the Sears-Haack body wave drag is presented in Eq. (12.46), where the Sears-Haack D/q is from Eq. (12.45).

The maximum cross-sectional area ($A_{\max}$) is determined from the aircraft volume-distribution plot. Inlet capture area should be subtracted from $A_{\max}$. The length term ℓ is the aircraft length except that any portion of the aircraft with a constant cross-sectional area should be subtracted from the length.

$$(D/q)_{\text{wave}} = E_{\text{WD}} \left[1 - 0.386(M - 1.2)^{0.57} \left(1 - \frac{\pi \Lambda_{\text{LE-deg}}^{0.77}}{100} \right) \right] (D/q)_{\text{Sears-Haack}} \quad (12.46)$$

" E_{WD} " is an empirical wave-drag efficiency factor and is the ratio between actual wave drag and the Sears-Haack value. For a perfect Sears-Haack body, $E_{\text{WD}} = 1.0$.

A very clean aircraft with a smooth volume distribution, such as a blended-delta-wing design, may have an E_{WD} as low as 1.2. A more typical supersonic fighter, bomber, or SST design has an E_{WD} of about 1.4–2.0. A poor supersonic design with a very bumpy volume distribution can have an E_{WD} of 2.0–3.0. The F-15, optimized for the dogfight instead of supersonic flight, has an E_{WD} of about 2.9 (Ref. 35).

Note, however, that this efficiency factor is less important in drag determination than the fineness ratio as represented by $(A_{\max}/\ell)$. This term is squared, which explains why area ruling that actually reduces $A_{\max}$ provides a far greater drag reduction than does merely smoothing the volume distribution without lowering $A_{\max}$.

Transonic Parasite Drag

The transonic flow regime extends roughly from Mach 0.8–1.2. The increase in drag as an aircraft accelerates through the transonic regime, called the "drag rise," is due to the formation of shocks, and is in fact the transonic portion of wave drag.

The critical Mach number (M_{cr}) occurs when shocks first form on the aircraft. The drag divergent Mach number (M_{DD}) is the Mach number at

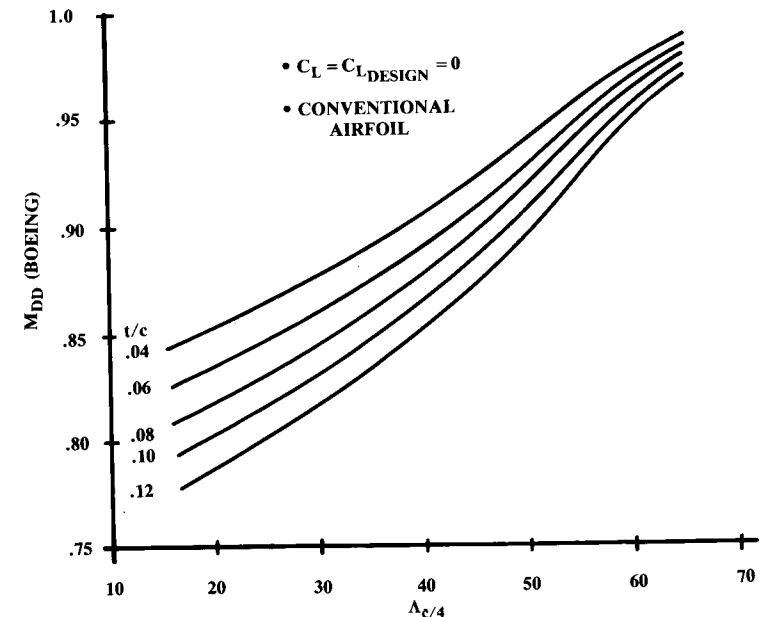


Fig. 12.26 Wing drag-divergence Mach number.

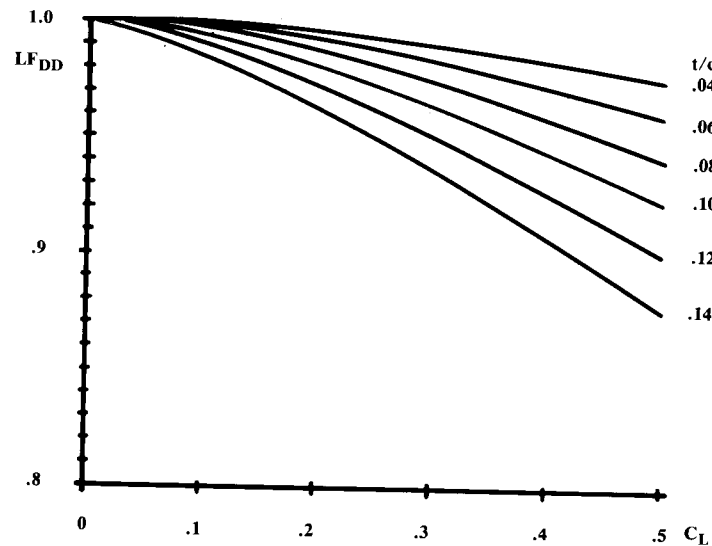


Fig. 12.27 Lift adjustment for M_{DD} .

which the formation of shocks begins to substantially affect the drag.

The definition of what speed constitutes M_{DD} is arbitrary, and several definitions are in use. The Boeing definition is that M_{DD} is where the drag rise reaches 20 counts. M_{DD} (Boeing) is usually about 0.08 Mach above the critical Mach number. The Douglas definition, also used by the Air Force in Ref. 37, is that M_{DD} is the Mach number at which the rate of change in parasite drag with Mach number (dC_{D_0}/dM) first reaches 0.10.

The Douglas M_{DD} is typically 0.06 Mach above the Boeing M_{DD} , and represents a drag rise of perhaps 80–100 counts. Jet transports usually cruise at about M_{DD} (Boeing), and have a maximum level speed of about M_{DD} (Douglas).

Shocks are formed on the top of the wing as a result of the increased airflow velocity, so M_{DD} reduces with an increased lift coefficient. For example, the Boeing 727 has an M_{DD} of about Mach 0.86 when the lift coefficient is only 0.1, but when the lift coefficient is increased to 0.3 the M_{DD} reduces to about Mach 0.82.

A preliminary estimate of wing M_{DD} (Boeing) is provided by Eq. (12.47) using Figs. 12.26 and 12.27. Figure 12.26 provides the wing drag divergence Mach number of an uncambered wing at zero lift. Figure 12.27 adjusts M_{DD} to the actual lift coefficient.

The last term in Eq. (12.47) is an adjustment for the wing design lift coefficient (i.e., camber and twist). Initially it can be assumed that the design lift coefficient is the same as the lift coefficient at cruise.

$$M_{DD} = M_{DDL=0} LF_{DD} - 0.05 C_{L_{\text{design}}} \quad (12.47)$$

If the wing uses a supercritical airfoil the actual thickness ratio should be multiplied by 0.6 before using these figures. This approximation is to account for the shock-delaying characteristics of the supercritical airfoil.

M_{DD} changes with lift coefficient. Lift coefficient changes with weight and altitude, both of which may change during cruise. To be completely accurate it is necessary to calculate M_{DD} for each point in the mission. For initial analysis, however, it is acceptable to use a single M_{DD} based upon a mid-mission weight and cruise altitude.

If the fuselage is relatively blunt it will experience shock formation before the wing does. In this case, M_{DD} is set by the shape of the forebody. Body M_{DD} can be estimated using Fig. 12.28 (Ref. 44), where “ L_n ” is the length from the nose to the longitudinal location at which the fuselage cross section becomes essentially constant. d is the body diameter at that location. If the fuselage is noncircular, d is an equivalent diameter based upon the fuselage cross-sectional area. Determine both wing and fuselage M_{DD} , and use the lower value.

The linear-wave-drag analysis gives completely incorrect results in the transonic regime. This analysis is called “linear” because the higher-order, nonlinear terms have been dropped from the aerodynamic equations to permit computation.

Some of these dropped nonlinear terms account for any changes in the airflow longitudinal velocity. At high supersonic speeds these terms have little effect compared to the far greater aircraft velocity.

However, the drag rise at transonic speeds is largely caused by the increase in airflow velocity over the top of the wing. Thus, drag rise below

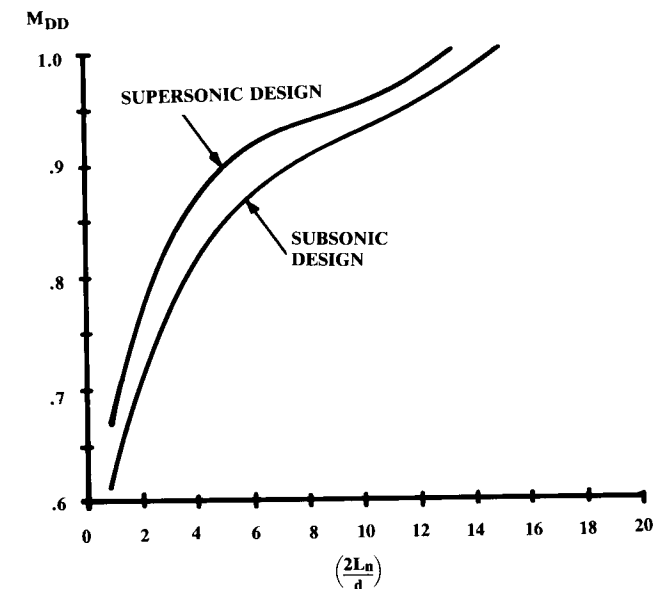


Fig. 12.28 Body drag-divergent Mach number.

Mach 1.0 is in fact caused by the terms that are dropped in the linear analysis! Transonic drag rise is therefore calculated to be zero by the linear-wave-drag methods. Only sophisticated nonlinear computational aerodynamic programs can give reasonable analytical results within the transonic regime.

Empirical methods for the calculation of the drag rise are presented in Ref. 37. These estimate the drag rise for the wing and fuselage separately, so the benefits of area ruling are ignored. These methods are very time-consuming and not very accurate, so an approximation technique is presented below.

For initial analysis the drag rise may be graphically estimated using a few rules of thumb, as shown in Fig. 12.29. The drag at and above Mach 1.2 (labeled A in the figure) is determined using Eq. (12.46) (divided by wing reference area). The drag at Mach 1.05 (labeled B) is typically equal to the drag at Mach 1.2.

The drag at Mach 1.0 (labeled C) is about half of the Mach 1.05 value. The drag rise at M_{DD} (determined above) is 0.002 by definition (labeled D). M_{cr} , the beginning of drag rise, is roughly 0.08 slower in Mach number than M_{DD} and is labeled E.

To complete the transonic-drag-rise curve from these points, draw a straight line through points B and C, extending almost to the horizontal axis. Then, draw a curve from M_{cr} through M_{DD} which fairs smoothly into the straight line as shown. If a smooth curve cannot be drawn, the M_{cr} point (E) should be moved until an approximately circular arc can be drawn. Finally, draw a smooth curve connecting B to A.

This crude technique may be used even for subsonic transport aircraft. The supersonic wave drag (point B) is determined from Eq. (12.46) although the aircraft will never fly at this speed. When calculating the Sears-

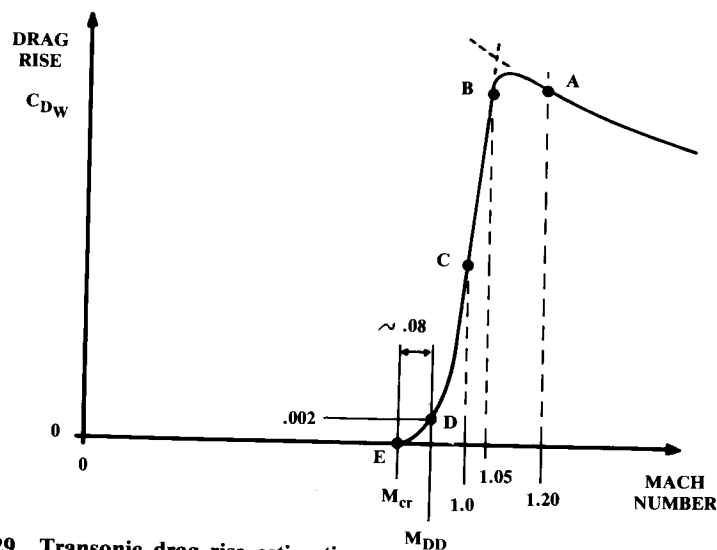


Fig. 12.29 Transonic drag rise estimation.

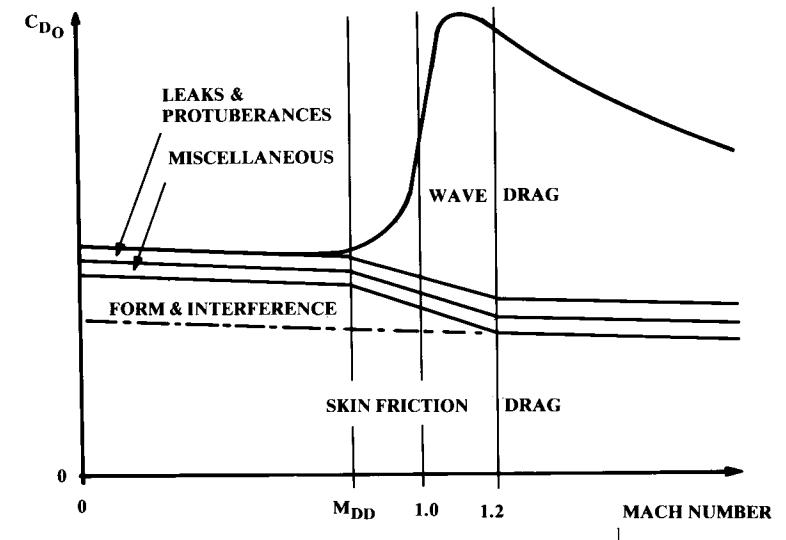


Fig. 12.30 Complete parasite drag vs Mach number.

Haack D/q for Eq. (12.46), remember to subtract from the aircraft length the portions of the aircraft where the cross-sectional area is constant.

Complete Parasite-Drag Buildup

Figure 12.30 illustrates the complete buildup of parasite drag vs Mach number for subsonic, transonic, and supersonic flight. The subsonic drag consists of the skin-friction drag including form factor and interference, plus miscellaneous drag and leak and protuberance drag. The supersonic drag includes the flat-plate supersonic skin-friction drag, miscellaneous drag, leak and protuberance drag, and wave drag.

In the transonic regime, the skin friction-drag is estimated simply by drawing a straight line between the skin-friction drag at M_{DD} (which includes form factor and interference) and the skin-friction drag at $M_{1.2}$ (which does not). This does not reflect any reduction in drag, merely a change in bookkeeping. The pressure drags represented by the form and interference terms at subsonic speeds are included in the wave-drag term at supersonic speeds.

In Fig. 12.31, the actual parasite drag and drag rise is shown for a number of aircraft.

12.6 DRAG DUE TO LIFT (INDUCED DRAG)

The induced-drag coefficient at moderate angles of attack is proportional to the square of the lift coefficient with a proportionality factor called the "drag-due-to-lift factor," or " K " [see Eq. (12.4)].

Two methods of estimating K will be presented. The first is the classical method based upon e , the Oswald span efficiency factor. Methods are pre-

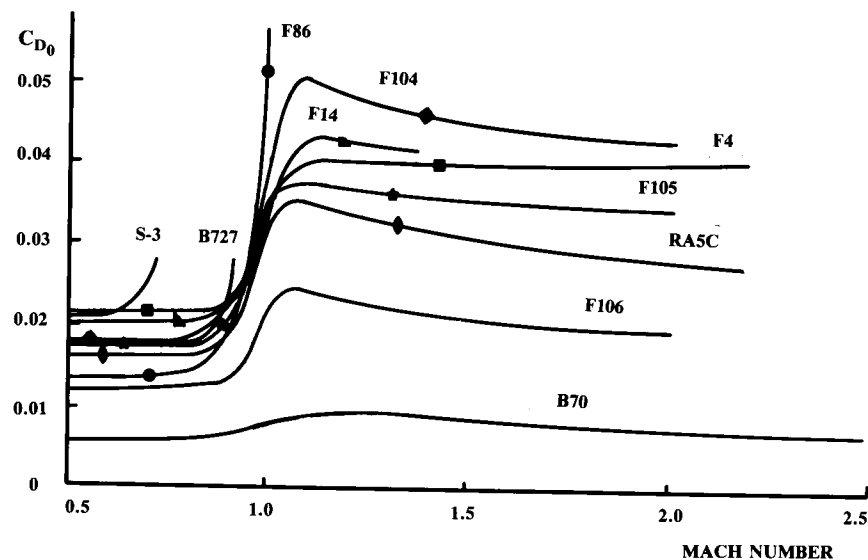


Fig. 12.31 Parasite drag and drag rise.

sented for subsonic monoplanes and biplanes along with an empirical equation for supersonic speeds.

The second method for the estimation of K is based upon the concept of leading-edge suction and provides, for high-speed designs, a better estimate of K , one that includes the effects of the change in viscous separation as lift coefficient is changed. This method also reflects the choice of the wing design lift coefficient on the drag due to lift at different lift coefficients.

Oswald Span Efficiency Method

According to classical wing theory, the induced-drag coefficient of a 3-D wing with an elliptical lift distribution equals the square of the lift coefficient divided by the product of aspect ratio and π . However, few wings actually have an elliptical lift distribution. Also, this doesn't take into account the wing separation drag.

The extra drag due to the nonelliptical lift distribution and the flow separation can be accounted for using e , the "Oswald span efficiency factor." This effectively reduces the aspect ratio, producing the following equation for K .

$$K = \frac{1}{\pi A e} \quad (12.48)$$

The Oswald efficiency factor is typically between 0.7 and 0.85. Numerous estimation methods for e have been developed over the years, such as those by Glauert and Weissinger. These tend to produce results higher than the e values of real aircraft. More realistic estimation equations based upon actual aircraft (Ref. 45) are presented below.

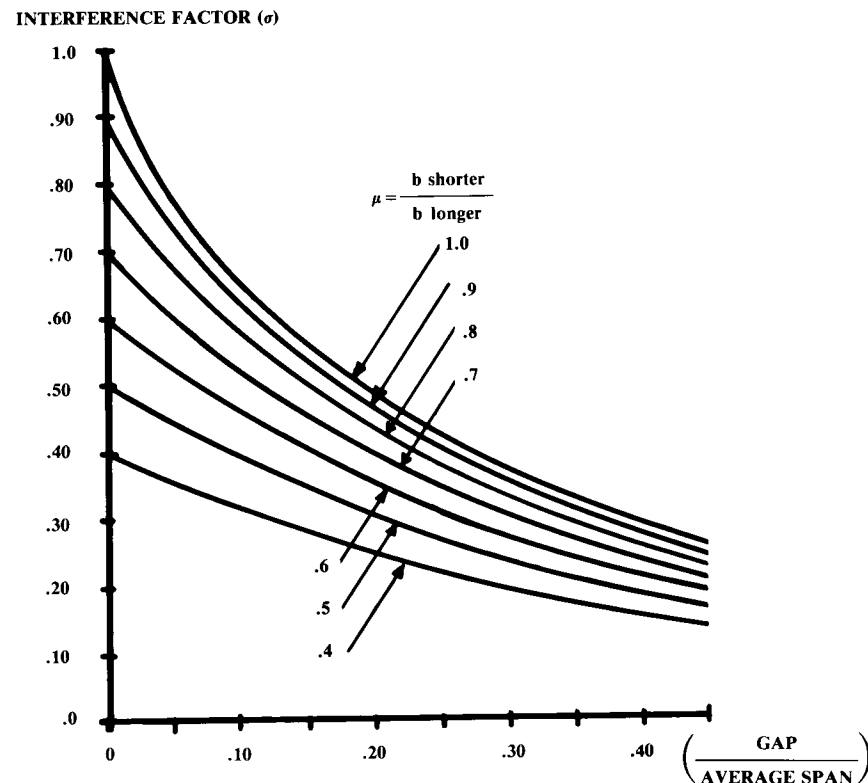


Fig. 12.32 Prandtl's biplane interference factor. (Ref. 12)

$$\text{Straight-Wing Aircraft: } e = 1.78(1 - 0.045A^{0.68}) - 0.64 \quad (12.49)$$

$$\text{Swept-Wing Aircraft: } e = 4.61(1 - 0.045A^{0.68})(\cos \Lambda_{LE})^{0.15} - 3.1 \quad (12.50)$$

$$(\Lambda_{LE} > 30 \text{ deg})$$

If the wing has end-plates or winglets, the effective aspect ratio from Eq. (12.10) or (12.11) should be used in Eq. (12.48).

Drag-due-to-lift for a biplane was first analytically determined by Max Munk in 1922, based upon the calculation of an equivalent monoplane span providing the same wing area and the same drag.

Prandtl developed an interference factor (σ , shown in Fig. 12.32) that is used in Eq. (12.51) to determine a biplane span efficiency factor (Ref. 12). The biplane aspect ratio is the square of the longer span divided by the total area of both wings.

$$\text{Biplane: } e = \frac{\mu^2(1 + r^2)}{\mu^2 + 2\sigma\mu r + r^2} \quad (12.51)$$

where

μ = shorter span/longer span

r = lift on shorter wing/lift on longer wing

(approximately = area of shorter wing/area of longer wing)

At supersonic speeds, the drag-due-to-lift factor (K) increases substantially. In terms of Oswald efficiency factor, e is reduced to approximately 0.3–0.5 at Mach 1.2. Equation (12.52) provides a quick estimate of K at supersonic speeds (Ref. 6), although the leading-edge suction method presented later is preferable.

$$\text{Supersonic Speeds: } K = \frac{A(M^2 - 1)}{4A\sqrt{M^2 - 1} - 2} \cos \Lambda_{LE} \quad (12.52)$$

Leading-Edge-Suction Method

Drag at angle of attack is strongly affected by viscous separation. At high lift-coefficients the drag polar breaks away from the parabolic shape represented by a fixed value of K in Eq. (12.4). The e method ignores this variation of K with lift coefficient. For a wing with a large leading-edge radius this is acceptable, but for most supersonic aircraft it gives a poor approximation.

A semi-empirical method for estimation of K allows for the variation of K with lift coefficient and Mach number; it is based upon the concept of “leading-edge suction.” Figure 12.33 illustrates the concept. The thick airfoil on the left is at an angle of attack below that at which substantial separation occurs. The flow streamlines curve rapidly to follow the leading-edge radius over the top of the wing.

This rapid curvature creates a pressure drop on the upper part of the leading edge. The reduced pressure exerts a suction force on the leading edge in a forward direction. This “leading-edge suction” force S is shown at the bottom of the figure in a direction perpendicular to the normal force N .

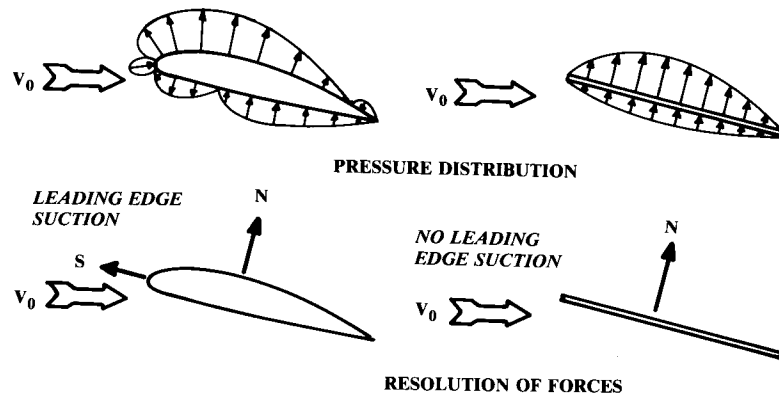


Fig. 12.33 Leading edge suction definition.

If there is no viscous separation or induced downwash, the leading-edge suction force exactly balances the rearward component of the normal force and the airfoil experiences zero drag. This is the ideal 2-D case described by d'Alembert's Paradox, and is called “100% leading-edge suction.”

A 3-D wing is considered to have 100% leading-edge section when the Oswald efficiency factor (e) exactly equals 1.0. When e equals 1.0, the induced-drag constant K exactly equals the inverse of the aspect ratio times π .

On the right side of Fig. 12.33 is a zero-thickness flat plate airfoil. Even without the leading-edge separation, which will almost certainly occur, this airfoil must have higher drag because there is no forward-facing area for the leading-edge pressure forces to act against. All pressure forces for a zero-thickness flat plate must act in a direction perpendicular to the plate, shown as N . There is zero leading-edge suction, and the lift and induced drag are simply N times the cosine or sine of the angle of attack [Eqs. (12.53) and (12.54)].

$$L = N \cos \alpha \quad (12.53)$$

$$D_i = N \sin \alpha = L \tan \alpha \quad (12.54)$$

or

$$C_{D_i} = C_L \tan \alpha \quad (12.55)$$

but (assuming α is small),

$$C_{D_i} = KC_L^2 \cong \alpha C_L \quad (12.56)$$

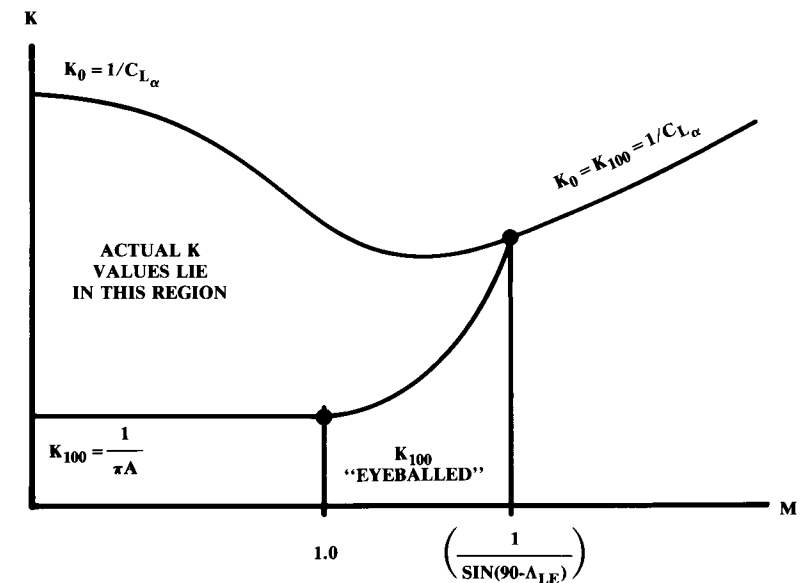


Fig. 12.34 0% and 100% K vs Mach number.

so,

$$K = \frac{\alpha C_L}{C_L^2} = \frac{\alpha}{C_L} = \frac{1}{C_{L\alpha}} \quad (12.57)$$

Thus, in the worst case of zero leading-edge suction, the drag-due-to-lift factor K is simply the inverse of the slope of the lift curve (in radians), as previously determined.

All real wings operate somewhere between 100 and 0% leading-edge suction. The percent of leading-edge suction a wing attains is called S (not to be confused with the force S in Fig. 12.33).

During subsonic cruise, a wing with moderate sweep and a large leading-edge radius will have S equal to about 0.85–0.95 (85–95% leading-edge suction). The wing of a supersonic fighter in a high- g turn may have an S approaching zero.

The method below for calculating K for high-speed aircraft is based upon an empirical estimate of the actual percent of leading-edge suction attainable by a wing, which is then applied to the calculated K values for 100 and 0% leading-edge suction. The actual K is calculated as a weighted average of the 100 and 0% K , as in Eq. (12.58).

$$K = SK_{100} + (1 - S)K_0 \quad (12.58)$$

The 0% K value is the inverse of the slope of the lift curve, as determined before. The 100% K value in subsonic flight is the inverse of the aspect ratio times π .

In transonic flight, the shock formation interferes with leading-edge suction. This increases the K value. When the leading-edge becomes supersonic, the suction goes to zero so the K value equals the 0% K value.

This occurs at the speed at which the Mach angle ($\arcsin 1/M$) equals the leading-edge sweep. Above that speed the wing has zero leading-edge suction so the K value is always the inverse of the slope of the lift curve.

For initial analysis, the supersonic behavior of the 100% K line may be approximated by a smooth curve, as shown in Fig. 12.34. This shows the typical behavior of the 100 and 0% K values vs Mach number.

The only unknown remaining is the value of S , the percent of leading-edge suction actually attained by the wing at the flight condition in question. S depends largely upon the leading-edge radius, and is also affected by the sweep and other geometric parameters.

S is also a strong function of the wing design lift coefficient and the actual lift coefficient. For any wing, the value of S is at a maximum when the wing is operating at the design lift coefficient. For most wings, S equals approximately 0.9 when operating at the design lift coefficient.

For a subsonic wing with large leading-edge radius and moderate sweep, the value of S will change very little with lift coefficient until the wing is near the stall angle of attack. For this reason, the induced drag for this type of wing can reasonably be estimated using the e method.

For the thin, swept wings typical in supersonic aircraft, the value of S can change substantially with lift coefficient. A wing with an S of 0.9 at its

design lift coefficient of 0.5 may have an S value less than 0.3 at a lift coefficient of 1.0.

Proper calculation of S for an actual wing is complex. An empirical approach may be used during conceptual design. Figure 12.35 provides a first-order estimate of the percent of leading-edge suction for a typical supersonic aircraft's wing, given the actual lift coefficient and the design lift coefficient (this determines which curve to use). Note that this chart assumes a well-designed wing, and at some later date the aerodynamics department must optimize the twist and camber to attain these values.

From Fig. 12.35 the leading-edge suction at various lift coefficients can be estimated. This allows adding curves to Fig. 12.34 that represent the estimated K value for different lift coefficients as a function of Mach number, as in Fig. 12.36. These are then used for total drag estimation via Eq. (12.4).

For the sake of comparison, Eqs. (12.59) and (12.60) relate S to e and ΔN (used in several other textbooks).

$$e = \frac{1}{(\pi A / C_{L\alpha})(1 - S) + S} \quad (12.59)$$

$$\Delta N = S \left(\frac{1}{C_{L\alpha}} - \frac{1}{\pi A} \right) \quad (12.60)$$

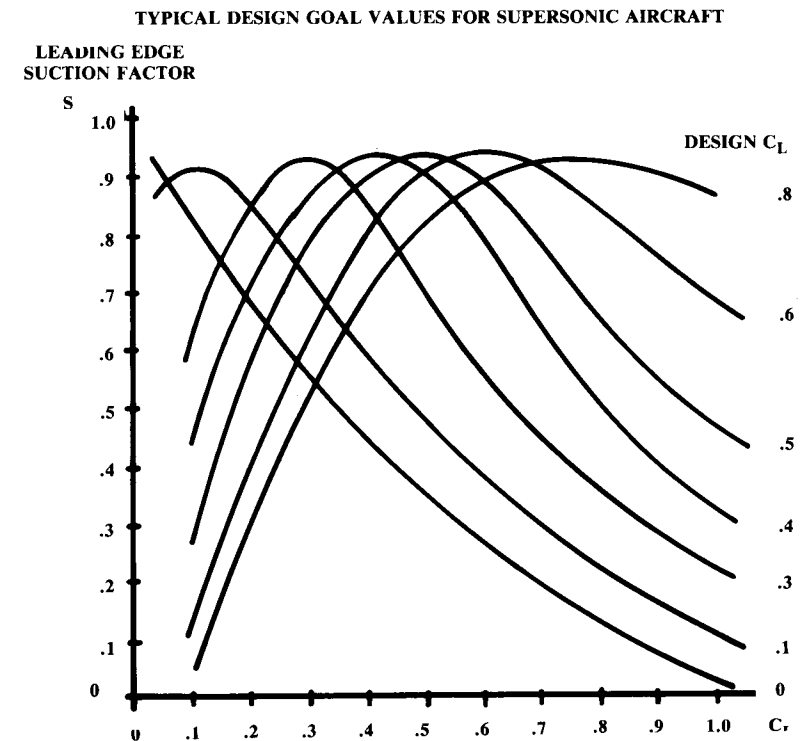


Fig. 12.35 Leading edge suction vs C_L .

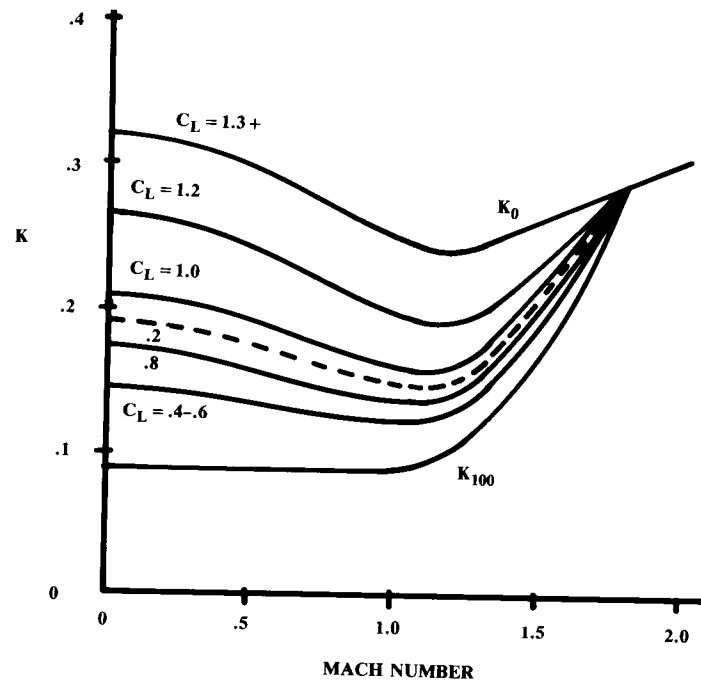


Fig. 12.36 Sample results— K vs Mach and C_L .

Ground Effect

When a wing is near the ground, say less than half the span away, the drag due to lift (K) can be substantially reduced. This is theoretically explained as a reduction in the induced downwash angle, but can be visualized as a trapping of a “cushion of air” under the wing. This effect is accounted for by multiplying K by the factor calculated in Eq. (12.61) (Ref. 70).

$$\frac{K_{\text{effective}}}{K} = \frac{33(h/b)^{1.5}}{1 + 33(h/b)^{1.5}} \quad (12.61)$$

where h is wing height above ground.

Trim Drag

The drag values used for performance calculations should include the trim drag. This additional induced drag is caused by the horizontal tail force required to balance (trim) the aircraft so that the total pitching moment about the aircraft c.g. will be zero for any given flight condition.

The tail usually trims the aircraft with a download that must be countered by additional lift from the wing. This produces an increase in the wing induced drag that must also be included in the trim drag.

Trim calculation is discussed in Chapter 16. The trim drag is determined using the above induced-drag methods once the tail lift force required for trim is known.

12.7 AERODYNAMIC CODES AND COMPUTATIONAL FLUID DYNAMICS (CFD)

Industry Practice for Aerodynamic Estimation

The aerodynamic methods presented above do not reflect current industry practice. Aircraft companies rely upon linearized computer codes such as the Harris wave-drag code, the Sommer and Short skin-friction code, and one of several panel codes such as USSAERO for induced effects. Newer panel codes such as PANAIR and QUADPAN are used to estimate the induced effects and the wave drag simultaneously and with better accuracy than the older codes.

These linearized computer codes can provide correct results only when the airflow around the aircraft is steady, unseparated, and does not contain any strong vortices. This is typically true only during cruising flight. Lift and drag at high angles of attack are estimated empirically using correlations to flight-test and wind-tunnel data for similar configurations.

The same is true for transonic lift and drag, where some of the very terms which are thrown away to linearize the equations are the longitudinal velocity-variation terms that produce the transonic shocks. Linearized wave-drag codes tend to over estimate the wave drag from Mach 1.0 to about Mach 1.2, and incorrectly predict zero drag rise below Mach 1.0. Empirical data is therefore used for the transonic regime.

Despite these problems, the standard industry practice of combining linearized computer codes with empirical data and corrections will produce good results in most cases. Actual flight-measured values of lift and drag are usually within about 2–10% of the estimates. Also, the estimates are the most accurate for the cruise portions of the flight where the most fuel is burned.

However, the fact that we can estimate a given design's lift and drag with reasonable accuracy does not guarantee that these methods will produce the best of all possible designs. Aerodynamic design has had to rely upon a trial-and-error process of design, analyze, test, and redesign.

Wind-tunnel testing offers a powerful tool for aircraft development. Unfortunately, the costs associated with detailed wind-tunnel testing tend to preclude an exhaustive evaluation of all possible designs. At a cost of several hundred thousand dollars per model, one is not likely to try something different just to see if it is better than the baseline design. Instead, the wind-tunnel is largely used to verify that a given design is workable.

It is sometimes difficult to identify the source of a problem during a wind-tunnel investigation because the wind tunnel “solves” all the flow equations simultaneously (i.e., viscous effects, vortex flow, induced effects, etc.). An unacceptable wiggle in the pitching-moment curve may be due to one of a number of causes, and the wind tunnel may not tell you which cause to fix!

Another problem with wind-tunnel testing is the Reynolds-number effect. Most wind tunnels cannot test at anything close to full-scale Reynolds numbers, resulting in substantial errors. Even worse, the optimal solution at a lower Reynolds number may not be the optimum at full-scale Reynolds

numbers. Who would propose a full-scale test on an airfoil or complete aircraft design that is known to be less-than-optimal in the wind tunnel?

CFD Definitions

It is for these reasons that Computational Fluid Dynamics (CFD) has rapidly become a key part of the aircraft design process. CFD is a catch-all phrase for a number of new computational techniques for aerodynamic analysis. It differs from prior aerodynamic codes by solving for the complete properties of the flowfield around the aircraft, rather than only on the surface of the aircraft.

CFD codes are based upon the Navier-Stokes (NS) equations, which were first derived in 1822. The NS equations completely describe the aerodynamics of a fluid (except for chemical-reaction effects at high temperatures).

NS includes equations based upon the existence of flow continuity, the conservation of momentum, and the conservation of energy. These are derived in many textbooks on theoretical and computational aerodynamics, and will not be repeated here.

The NS equations seem straightforward enough but cannot be analytically solved for any useful flow conditions. The author of Ref. 80 describes them as "some of the nastiest differential equations in theoretical physics."

The history of theoretical aerodynamics to date can largely be described as the quest for solvable simplifications of the NS equations. The classical lifting-line theory is one such simplification, as are the linearized wave-drag and panel codes, the Euler Codes, and the various NS codes.

There is a complete hierarchy of aerodynamic codes depending upon how many flow phenomena are neglected from the full NS equations. No current codes attempt to actually solve the true, full NS equations, due to the difficulty in mathematically analyzing turbulence. Turbulence occurs at the molecular level, which would probably require gridding the flowfield with billions of molecule-sized grids.

The current so-called "Navier-Stokes Codes" actually use a simplification in the handling of turbulence, which is the most difficult flow phenomena to analyze mathematically. Turbulence is handled with some type of separate statistically-calibrated model apart from the NS solution.

The most sophisticated codes to date, the "Large Eddy Simulation" codes, use a statistically-based turbulence model for small-scale turbulence effects. Large Eddy codes are capable of directly analyzing the larger turbulent eddies. The Large Eddy Simulation is beyond the capabilities of current computers for a complex aircraft configuration, but has been used for simplified geometries.

The current state of the art for complex aircraft configurations, the "Reynolds-Averaged Navier-Stokes," has both large and small eddies (turbulence) modeled statistically. Reynolds-Averaged codes can handle most of the complex flow phenomena that elude linearized codes, including vortex formation, separation, transonic effects, and unsteady effects.

Reynolds-Averaged codes are being used on the National Aerospace Plane (NASP) project to solve particular design problems where no other methods can give correct results. Unfortunately, Reynolds-Averaged codes are extremely expensive to set up and run. One recent example took 20 hours

on a Cray XMP-22 to yield results at one Mach number, altitude, angle of attack, and angle of sideslip. Because of the expense these codes are not yet useful for routine design work.

The NS simplification emerging as the workhorse for design analysis, the "Parabolized Navier-Stokes" (PNS), drops the viscous terms in the streamwise direction, which ignores streamwise separation effects. However, with a good turbulence model the PNS codes give correct and illuminating results for most design problems.

If all viscosity effects are ignored and the flow is assumed to be steady, the Euler equations are derived from the NS equations. Euler codes are much cheaper to run than even PNS codes, and are widely used at this time. The Euler codes can handle vortex formation, and with the addition of a separate boundary-layer code, can also realistically estimate viscous and separation effects.

The "Potential Flow" equations are further simplified from the Euler equations by dropping the rotational terms. This prevents the analysis of vortex flow, which is important at high angles of attack but is of lesser importance during cruise conditions. Potential Flow codes can handle transonic shock formation and are very useful for transonic design compared to the linearized methods. The Potential Flow codes are not usually considered to be true "CFD," but are probably the most widely-used aerodynamic codes that treat the entire flowfield rather than just the surface conditions.

The "Linearized" aerodynamic codes are based upon a further simplification to the Potential Flow equations by neglecting the higher-order terms. It is assumed that, since they involve small quantities multiplied by other small quantities, they must be very small and therefore negligible. At transonic speeds, however, these terms are not so small!

The Linearized Potential Flow equations are the basis of the standard industry methods described at the beginning of this section. These include the Harris Wave Drag and the USSAERO and similar panel methods. With further simplifications, such classical methods as the lifting-line theory are derived.

To recap, only the Large Eddy, Reynolds-Averaged, and PNS codes are considered to be true "Navier-Stokes" codes. However, the Euler, Potential Flow, and Linearized aerodynamic codes are in fact successive simplifications of the NS equations. The choice of code for a given design problem depends upon the nature of the problem and the available budget (and not always in that order!).

Applications of CFD

CFD does not replace the wind tunnel. In fact, it really doesn't even reduce the number of wind-tunnel test hours. CFD does permit you to design a better airplane by a truer understanding of the flow field around it. Not only do the CFD codes determine the entire flowfield around the aircraft, but also, unlike the wind tunnel, the flowfield determination is done at the full-scale Reynolds number.

A perfect example of the use of CFD can be found at every major commercial airport in the country. The installation of the fuel-efficient CFM-56

engine on the Boeing 737 would not have been possible without the use of CFD, as described in Ref. 80.

The original Boeing 737 used the P&W JT8D, a low-bypass-ratio engine that was mounted in a wing-conformal nacelle. The nacelle barely cleared the ground, providing a minimum-weight landing gear.

When Boeing decided to develop an updated version of the 737, the CFM-56 engine was the logical choice as a modern fuel-efficient engine of the required thrust class. However, it has a diameter some 20% greater than the old engine. Furthermore, the CFM-56 exits its fan air up front like most modern turbofans. For this reason, a wing-conformal nacelle was not possible.

In a prior chapter, the cited rough rule of thumb for podded jet nacelles said that the inlet should be about two inlet diameters forward of the wing and about one inlet diameter below it. A more-refined empirical method of locating a turbofan engine indicated that the geometry shown in Fig. 12.37a was the closest acceptable nacelle spacing. Clearly this posed a ground clearance problem!

The empirical rules for nacelle spacing were based upon years of trial and error in the wind tunnel. Closer spacings were found to increase cruise drag, although the wind-tunnel investigations had not clearly determined just exactly what this "interference" drag consisted of. Various suspects included increased skin friction due to supersonic velocity, increased separation, shock effects, and a change in the wing's spanwise lift distribution resulting in an increase in the induced drag.

Through the use of a nonlinear potential flow panel program (CFD state of the art in the 1970's), Boeing was able to determine that it was in fact the induced drag effect that was creating the "interference drag." This important piece of information had not been determined in 20 years of wind-tunnel testing!

With this information, Boeing was then able to contour a closely-spaced podded nacelle to prevent any change in the wing's spanwise lift distribution. This was possible with CFD because the entire flowfield is solved, allowing the designers to study the streamlines and pressure fields resulting from various design changes. The designers sought to minimize the impact of the nacelle on the streamlines of the bare wing.

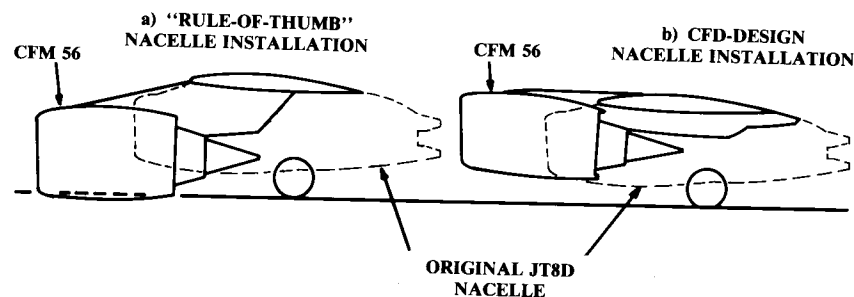


Fig. 12.37 CFD example: Boeing 737 nacelle (after R. Bengelink, AIAA Paper 88-2043).

Figure 12.37b shows the result, namely, a nacelle of extremely close spacing to the wing that, nevertheless, has acceptable drag characteristics.

The formation and effects of vortices at high angles of attack represent another area of substantial concern to the designers of fighter aircraft. These vortices produce completely different lift, drag, and pitching-moment characteristics than those which would be predicted using the linear methods.

Reference 81 details the CFD solution of a typical vortex-flow problem using a Lockheed Euler code called TEAM (Three-Dimensional Euler Aerodynamic Method). Figure 12.38 (reprinted with permission) shows the close match between the calculated and the measured pressures over the double-delta configuration used in the study. The vortex region can be seen by the diagonal pressure contours in both calculated and measured plots.

Figure 12.39 illustrates the power of CFD to fully analyze the flowfield around the aircraft rather than just at the surface. Arrows are used to depict the flowfield at four longitudinal locations on the aircraft. The length of the arrows shows the relative magnitudes of the flow velocity in the plane of the cross section. The vortex flow is clearly seen beginning inboard toward the

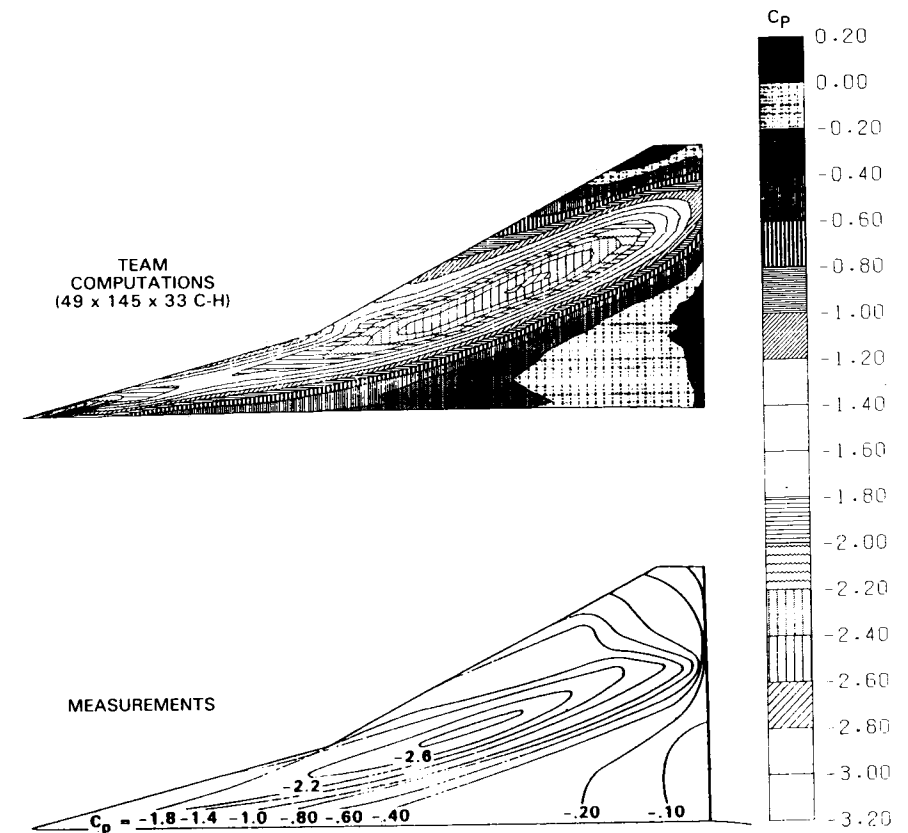


Fig. 12.38 Correlation of computed and measured surface pressure contours; 75°/62° double-Delta wing body; $M_\infty = 0.3$; $\alpha = 20^\circ$. (Ref. 81)

front of the wing, and growing and moving outward toward the rear of the aircraft.

CFD Issues and Challenges

We have come a long ways since 1879 when the annual proceedings of the British Royal Aeronautical Society could say, “Mathematics up to the present day has been quite useless to us in regard to flying” (quoted in Ref. 80). However, there are still many problems associated with the use of CFD

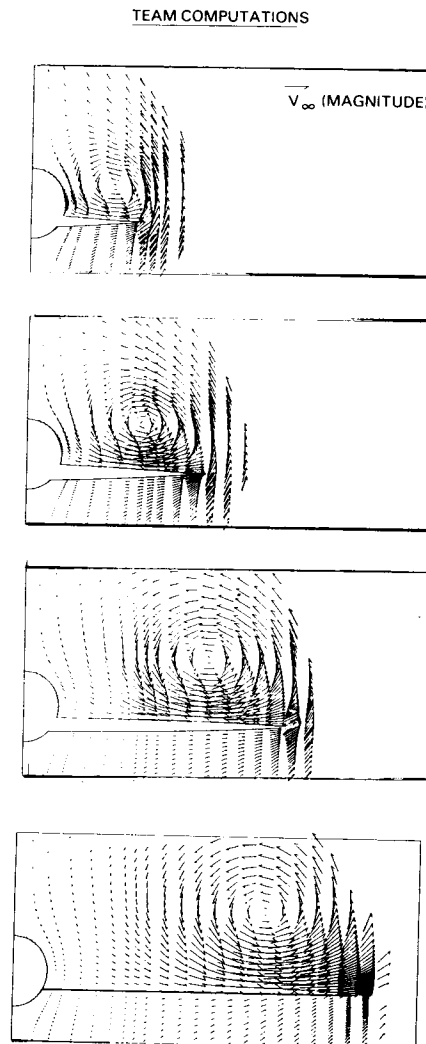


Fig. 12.39 Correlation of computed and measured cross-plate velocity fields; $75^\circ/62^\circ$ double-Delta wing body; $M_\infty = 0.3$; $\alpha = 20^\circ$. (Ref. 81)

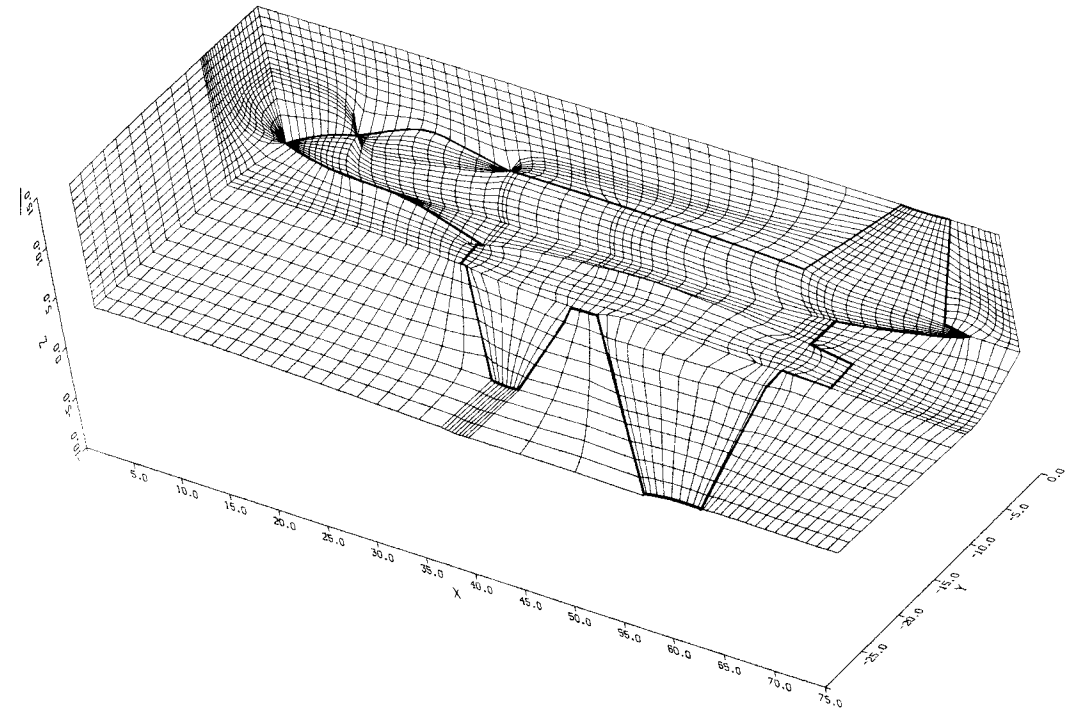


Fig. 12.40 Flowfield gridding. (Ref. 82)

for routinely solving aircraft design problems. Two problems are especially important: the influence of the turbulence model and the requirements for flow gridding.

The use of separate turbulence models for NS codes has been discussed. The results of the various NS codes are very sensitive to the turbulence model used, especially when separated flow is present.

CFD codes tend to produce reasonable-looking flowfields and pressures, but sometimes the integration of the calculated pressures yields lifts, drags, and moments which do not match experience. Reproduction of experimental data sometimes requires extensive “calibration” (i.e., fudging!) of the turbulence model. For this reason, CFD results are always somewhat suspect until the code has been checked against experimental data for a similar configuration.

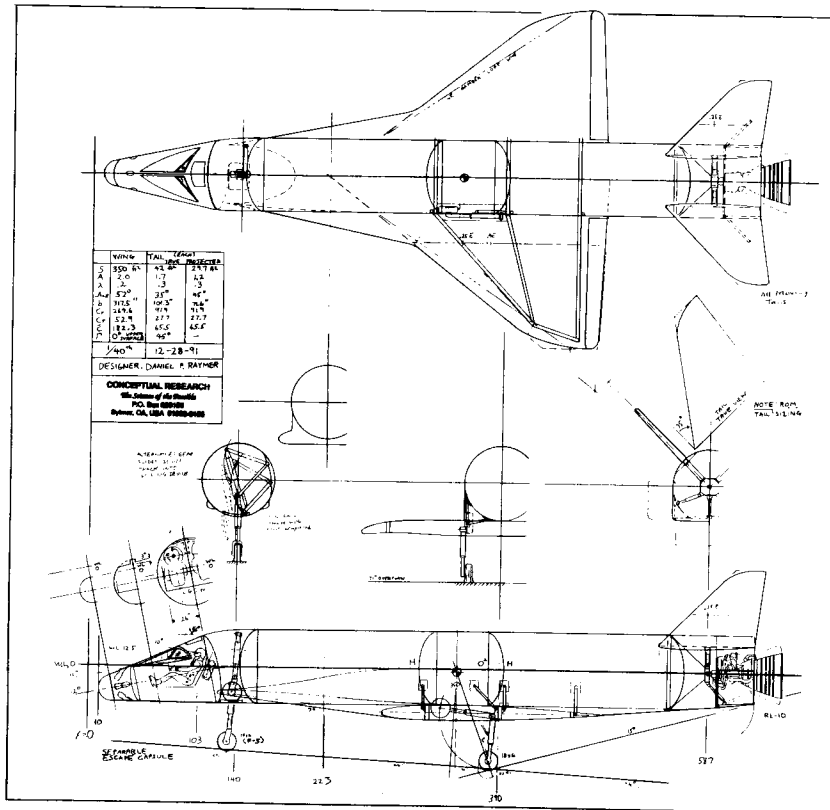
The need to grid the entire flowfield around the aircraft presents another big problem for CFD users. “Gridding” refers to the breaking up of the space around the aircraft into numerous small blocks, or “cells,” usually of roughly hexahedral shape. CFD methods calculate the flow properties

within each cell, using various convergence schemes to equate the flow properties along the boundaries connecting the cells.

While gridding the space around a simple cylinder or a lone wing can be easily automated, the gridding of the flowfield around a full aircraft must currently be done manually and can literally take months. Figure 12.40 (reprinted with permission from Ref. 82) illustrates the complexity of the flowfield gridding. Note, for example, where the canopy meets the fuselage and where the cells must fan out in the empty region between wing and canard.

Gridding is especially important because the CFD results are highly sensitive to the shaping of the cells. You can actually get different answers for the same aircraft using two different gridding schemes. According to the author of Ref. 82, “this sensitivity is more pronounced than that due to the type of mathematical model being used, e.g., NS vs Euler equations.”

To address the gridding problem, researchers are investigating artificial intelligence (AI) approaches to gridding. Another approach is the computationally-adaptive gridding in which the gridding scheme is automatically adjusted based upon the CFD results.



Skyrocket Aerospace Plane Concept

13 PROPULSION

13.1 INTRODUCTION

All forms of aircraft propulsion develop thrust by pushing air (or hot gases) backward. In a simplified case the force obtained can be determined using Newton's equation ($F = ma$) by summing all the accelerations imparted to the air.

This is shown for fluid flow in Eq. (13.1), where s is the cross-sectional area of the fluid acted upon by the propulsion system, V is the fluid velocity, and the zero subscript indicates the freestream condition. The propeller or jet engine is assumed to “magically” accelerate the air from velocity V_0 to V .

The rate of useful work done by the propulsion system, called the “thrust power (P_t),” equals the product of the thrust force and the aircraft velocity [Eq. (13.2)].

The kinetic energy imparted to the fluid by the propulsion system is determined by the difference in fluid velocity, as shown in Eq. (13.3).

The “propulsion efficiency (η_{PE})” is defined as the ratio of thrust power obtained to energy expended, as shown in Eq. (13.4). Note that the efficiency is maximized when there is no change in fluid velocity. Unfortunately, at this condition Eq. (13.1) shows that the thrust is zero!

$$F = ma = \dot{m}\Delta V = (\rho Vs)(V - V_0) = \rho sV(V - V_0) \quad (13.1)$$

$$P_t = FV_0 = \rho sV(V - V_0)V_0 \quad (13.2)$$

$$\Delta E = \frac{1}{2}\dot{m}V^2 - \frac{1}{2}\dot{m}V_0^2 = \frac{1}{2}\rho Vs(V^2 - V_0^2) = \frac{\rho s}{2}V(V^2 - V_0^2) \quad (13.3)$$

$$\eta_{PE} = \frac{P_t}{\Delta E} = \frac{2}{V/V_0 + 1} \quad (13.4)$$

There is an unavoidable tradeoff between thrust and efficiency as determined by the ratio between exhaust and freestream fluid velocity. For maximum thrust, this ratio must be very high. For maximum efficiency, this ratio should be unity.

If the exhaust velocity is reduced to little more than the freestream velocity for improved efficiency, the cross-sectional area of fluid affected by the

propulsion system must approach infinity to maintain constant thrust. A typical turbojet will operate with the ratio of exhaust velocity to freestream velocity at well above 3.0, whereas a typical propeller aircraft will operate with this ratio at about 1.5.

The analysis above is too simplistic for actual thrust calculation. It falsely assumes that the fluid velocity is constant throughout the exhaust and that all of the accelerations experienced by the airmass occur at the propeller plane or within the jet engine.

Actually, the exhaust of a jet engine is usually at a higher pressure than the outside air, so the flow expands after leaving the nozzle. In other words, the air is still accelerating after the aircraft has passed.

For a propeller, the airmass acceleration doesn't even occur at the propeller disk! Roughly half the airmass acceleration occurs before reaching the propeller, and the other half occurs after passing the propeller.

Propulsion force estimation is also complicated by the fact that the propeller flowfield or jet intake and exhaust will influence the whole flowfield of the aircraft. It has already been mentioned that a pusher propeller will reduce the drag of a stubby aft fuselage by "sucking" air inward and preventing flow separation. Should this reduced drag be considered a part of the propulsive force because it is controlled with the throttle? What about the increased drag due to the propeller wake on a conventional airplane?

For a propeller aircraft, most of the propulsive force is exerted directly on the aircraft by the pull (or push) of the propeller itself through the propeller shaft. The propeller shaft is usually connected to the engine so the engine mounts actually pull (or push) the rest of the aircraft through the air.

For a jet aircraft the force exerted through the engine mounts may only be a third of the total propulsive force. Figure 13.1 shows the thrust contributors for a typical Mach 2.2 nacelle. The engine itself only contributes about 8% of the total. The nozzle, which generates thrust by expanding the high-pressure engine exhaust, contributes almost 30%.

The inlet system uses a system of shocks to slow the air to a subsonic speed. This creates a substantial drag. However, the subsonic expansion inside the inlet duct contributes a positive force that more than makes up

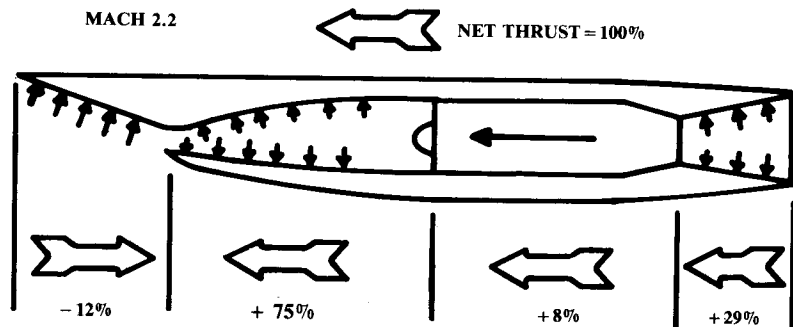


Fig. 13.1 Turbojet thrust contributors.

for the external inlet system drag. In fact, the subsonic expansion inside the inlet duct is the largest single contributor of thrust! This illustrates the difficulty of calculating thrust by any simple model.

This chapter provides methods for estimating the net thrust provided by a propeller or jet engine as a part of the overall vehicle analysis and optimization. These methods are simplified to permit experiencing the whole design process within the time devoted to the typical college design course. The chapter will also introduce the vastly more complicated process of installed-thrust estimation used at major aerospace firms. Reference 46 (highly recommended) provides a detailed treatment of jet-engine design and installation.

13.2 JET-ENGINE THRUST CONSIDERATIONS

Before we begin the discussion of jet-engine installed thrust, a brief introduction to jet-engine cycle analysis and its effect on design is in order. As mentioned in Chapter 10 and shown in Fig. 10.1, a jet engine develops thrust by taking in air, compressing it (via the inlet duct and the compressor), mixing in fuel, burning the mixture, and expanding and accelerating the resulting high-pressure, high-temperature gases out the rear through a nozzle.

To provide power to drive the compressor, a turbine is placed in the exhaust stream which extracts mechanical power from the high-pressure gases. If greater thrust is required for a short period of time, an afterburner can be placed downstream of the turbine permitting the unburned air in the turbine exhaust to combust with additional fuel and thereby increase the exhaust velocity.

"Gross thrust" is produced as a result of the total momentum in the high-velocity exhaust stream. "Net thrust" is calculated as the gross thrust minus the "ram drag," which is the total momentum in the inlet stream. Note that the ram drag, which results from the deceleration of the air taken into the inlet, is included in the engine cycle analysis performed by the engine manufacturer to determine net uninstalled thrust.

Jet-engine cycle analysis, as detailed in Ref. 46 and other propulsion texts, is the straightforward application of the laws of thermodynamics to this Brayton engine cycle. In an "ideal" analysis, the efficiencies of components such as compressors and turbines are assumed to be 100%, i.e., no losses, and the resulting thrust for a given fuel flow, altitude, and Mach number is calculated. While optimistic, such ideal cycle analysis produces results in the "right ballpark" and illustrates the trends produced by varying such cycle parameters as overall pressure ratio, turbine inlet temperature, bypass ratio, and flight condition. These are discussed below.

One over-riding factor in the determination of jet-engine performance is that the net thrust produced is roughly proportional to the air mass flow (velocity $\times$ air density $\times$ airflow cross section) entering the engine. For a modern afterburning turbojet engine, roughly 126 pounds of thrust (the "specific net thrust") is developed for each pound per second of air taken in by the engine. For a turbofan engine, a specific net thrust of roughly 40–60 can be obtained.

An increase in air density such as at low altitude or low outside air temperature would therefore increase thrust by increasing mass flow. Hot day takeoffs from a high-elevation airport such as Denver pose problems because the reduction in air density causes a reduction in mass flow, and hence, thrust.

Similarly, an increase in aircraft velocity also increases thrust due to ram effect increasing the mass flow. However, for a typical subsonic jet, the exhaust comes out the nozzle at a choked condition, and so the exit velocity equals the speed of sound regardless of aircraft velocity. As aircraft velocity approaches the speed of sound, the thrust is therefore reduced for the choked exit nozzle [see Eq. (13.1)]. When combined with the favorable ram effect, this results in a relatively constant thrust as velocity increases for the typical subsonic jet, dropping off as transonic speeds are reached.

For supersonic jet engines, a variable area, converging-diverging nozzle is typically employed which permits supersonic exhaust velocities. Therefore, the ram effect does cause the thrust to tend to increase with increasing velocity until at high Mach numbers where excessive total pressure losses occur in the inlet, resulting in thrust degradation. The Mach number at which inlet losses become excessive is determined by the number of shocks and the extent of variable geometry employed, as described in Chapter 10.

Thrust and propulsive efficiency are strongly affected by the engine's overall pressure ratio (OPR). OPR is the ratio of the pressures at the engine exhaust plane and inlet front face. This pressure ratio is a measure of the engine's ability to accelerate the exhaust, which produces thrust. OPRs usually range from about 15 to 1 to about 30 to 1.

Another key parameter which currently limits turbine engine performance is the turbine inlet temperature (TIT). As mentioned earlier, it would be desirable for maximum thrust and efficiency to combust at the stoichiometric air-fuel ratio of about 15 to 1. This produces temperatures far too high for current turbine materials, even using the best available cooling techniques. Instead, a "lean" mixture of about 60 to 1 (air to fuel) is used, with the extra air holding down the TIT to about 2000–2500°F. This results in less thrust and thermal efficiency, and so a key objective in propulsion technology development has always been the increase in allowable TIT.

To increase propulsive efficiency, the turbofan engine uses an oversized fan with some of the accelerated fan air "bypassed" around the engine, not being used for combustion. This has the effect of allowing the engine to accelerate a larger cross-sectional area of air by a smaller change in velocity, which increases efficiency as determined by Eqs. (13.1) and (13.4). The bypass ratio was defined in Chapter 10 as the ratio of the mass flows of the bypassed air and the air that goes through the core of the engine to be used for combustion.

A higher bypass ratio, which enables the engine to accelerate a larger cross section of air, produces higher efficiency and hence greater thrust for a given expenditure of fuel. However, the fan alone cannot efficiently accelerate the air to transonic or supersonic exit speeds, and so this favorable effect works only at lower speeds. As was shown in Fig. 10.2, the high-bypass turbofan is best at subsonic speeds, giving way to the low-bypass ratio turbofan at the low supersonic speeds. At higher supersonic speeds, say over about Mach 2.2, the pure turbojet is superior.

13.3 TURBOJET INSTALLED THRUST

Chapter 10 described statistical methods for estimating installed thrust and specific fuel consumption for jet engines. These are suitable for initial sizing and performance estimation. For a more sophisticated analysis it is necessary to estimate analytically the installed performance of an existing engine or a proposed new engine from the uninstalled engine data.

Uninstalled engine data can be obtained from the engine manufacturer. Data for several conceptual engines is summarized in the appendices. For a new-design engine, the data must be estimated using cycle-analysis equations such as in Ref. 46. Usually the designers in an aircraft company will obtain the uninstalled-engine data for a proposed engine from the engine manufacturer, and then will correct the data for installation effects.

It is common early in design studies to approximate the performance of a new-design engine by a "fudge-factor" approach. An existing engine with approximately the same bypass ratio is selected, and its size, weight, and performance data are multiplied by factors based upon the expected improvements by applying advanced technologies.

For example, it might be assumed that an engine designed ten years from now would have 25%-less specific fuel consumption, 30%-less length, and 30%-less weight compared to an existing engine. Such "fudge-factors" are based upon either historical trend analysis or an approximate cycle analysis for expected technology improvements.

It is assumed below that uninstalled-engine data is available, either from an engine manufacturer, a preliminary cycle analysis, or a fudge-factor approach based upon some given engine (such as those in the appendices).

13.4 THRUST-DRAG BOOKKEEPING

Bookkeeping is not normally considered an engineering subject. However, the interactions between thrust and drag are so complex that only a bookkeeping-like approach can assure that all forces have been counted once and only once. It is not at all uncommon to discover, halfway through an aircraft design project, that some minor drag item has been either included in both drag and thrust calculations or has been ignored by both departments under the assumption that it is being included by the other!

Each aircraft company develops its own system for thrust-drag bookkeeping. In fact, a different system may be developed for each design project. In most cases the guiding principle for determining whether a force is considered a part of the drag or the thrust is whether that force changes when the throttle setting is changed.

In an afterburning jet engine, for example, the nozzles open wide when the throttle is advanced to the afterburning position. This changes the aerodynamic drag on the outside of the nozzles, so the nozzle aerodynamic drag is counted as a reduction in the engine thrust in many thrust-drag bookkeeping systems.

In other thrust-drag bookkeeping systems, the nozzle drag is separated into two components: (1) the drag value at some fixed nozzle setting (usually full open), which is included in the aerodynamic drag calculation, and

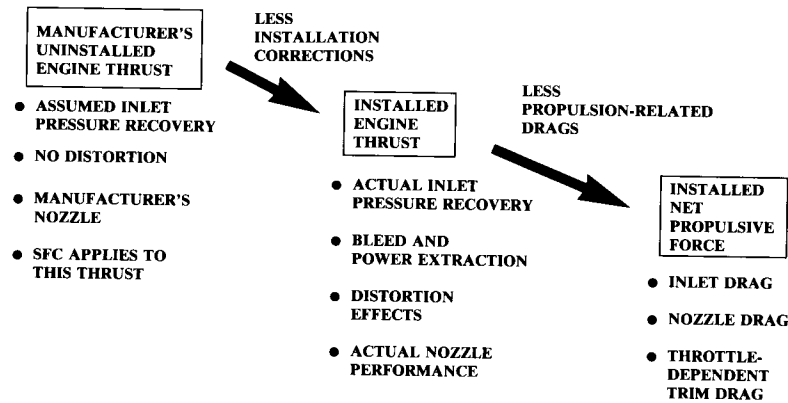


Fig. 13.2 Installed thrust methodology.

(2) the variation of drag as the nozzle setting is changed, which is included in the propulsion-installation calculations.

Either bookkeeping approach will give correct results providing that the Aerodynamics and Propulsion departments both understand it.

Thrust-drag bookkeeping becomes especially complex when sorting out the results of wind-tunnel testing. Different wind-tunnel models are used to test different thrust and drag items. The model used for determining basic aerodynamic and stability derivatives is usually unpowered, and a separate powered model is used to estimate propulsion effects. Lack of a mutually understood bookkeeping system by both the Aerodynamic and Propulsion departments will cause chaos.

The student should realize that the organization of this book assumes a thrust-drag bookkeeping system. Items presented in this chapter as reductions to thrust may be considered to be drag items in another bookkeeping system. Reference 35 contains a detailed review of the subject of thrust-drag bookkeeping.

13.5 INSTALLED-THRUST METHODOLOGY

The actual available thrust used in performance calculations—called the “installed net propulsive force”—is the uninstalled thrust corrected for installation effects, minus the drag contributions that are assigned to the propulsion system by the selected thrust-drag bookkeeping system. This is depicted in Fig. 13.2.

The “manufacturer’s uninstalled engine thrust” is obtained by cycle analysis and/or testing, or can be approximated using fudge-factors as described above. The manufacturer’s engine data is based upon some assumed schedule of inlet pressure recovery vs Mach number. Inlet pressure recovery (P_1/P_0) is the total pressure at the engine front face (1) divided by the total pressure in the freestream (0).

For subsonic engines, it is frequently assumed that the inlet pressure recovery is perfect, i.e., $P_1/P_0 = 1.0$. For supersonic military aircraft a Mil-Spec formula is used. Inlet distortion, engine bleed, and engine power extraction are usually assumed by the manufacturer to be zero. Also, the engine data is based upon the manufacturer’s nozzle design.

Note that the SFC values supplied with the engine are based upon this uninstalled-engine thrust, not the installed net propulsive force. When determining fuel usage, the SFC values must be ratioed up by the ratio between uninstalled-engine thrust and the net propulsive force (i.e., the thrust required for the desired performance).

The “installed-engine thrust” is the actual thrust generated by the engine when installed in the aircraft. This is obtained by correcting the thrust for the actual inlet pressure recovery and nozzle performance, and applying thrust losses to account for engine bleed and power extraction.

“Inlet distortion” refers to pressure and velocity variations in the air delivered to the engine. It primarily affects the allowable operating envelope of the engine.

The installed net propulsive force is the installed engine thrust minus the inlet, nozzle, and throttle-dependent trim drags. The steps depicted in Fig. 13.2 are detailed below.

Installed Engine Thrust Corrections

The manufacturer’s uninstalled engine thrust is based upon an assumed inlet pressure-recovery. For a subsonic engine, it is typically assumed that the inlet has perfect recovery, i.e., 1.0. Supersonic military aircraft engines are usually defined using an inlet pressure-recovery of 1.0 at subsonic speeds and the inlet recovery of Eq. (13.5) (MIL-E-5008B) at supersonic speeds. Added to this is the pressure recovery loss due to internal flow in the inlet duct itself. Typically this adds 2–3% to the losses.

Figure 13.3 shows this reference inlet pressure-recovery plotted vs Mach number, compared to the recovery available for a normal-shock inlet and external compression inlets with one, two, and three ramps.

$$\left(\frac{P_1}{P_0}\right)_{\text{ref}} = 1 - 0.075(M_\infty - 1)^{1.35} \quad (13.5)$$

The external compression inlets of Fig. 13.3 are of movable ramp design with a perfectly optimized schedule of ramp angles as a function of Mach number. To determine the pressure recovery of a fixed or less-than-perfect inlet, the shock tables in the appendix should be used.

The pressure losses inside the inlet duct must also be accounted for. These losses are determined by the length and diameter of the duct, the presence of bends in the duct, and the internal Mach number.

For initial evaluation of a typical inlet duct, an internal pressure recovery of 0.96 for a straight duct and 0.94 for an S duct may be used. The short

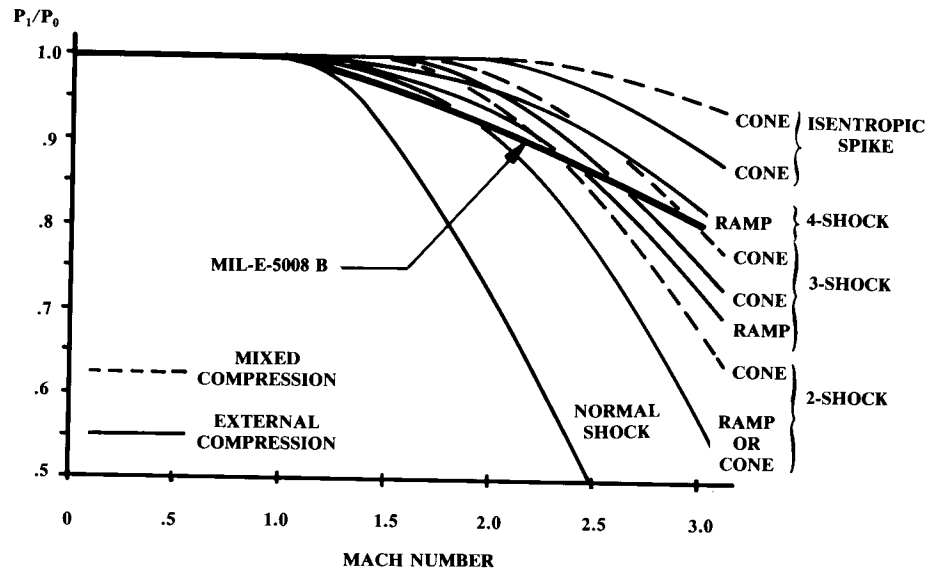


Fig. 13.3 Reference and available inlet pressure recovery.

duct of a subsonic podded nacelle will have a pressure recovery of 0.98 or better. More detailed estimation of inlet internal-pressure loss is based upon experimental data (see Ref. 27), and requires a separate evaluation at each Mach number.

Figure 13.4 provides the actual inlet pressure recoveries of some existing designs. This figure may be used for pressure-recovery estimation during early design studies.

Reducing inlet pressure recovery has a greater-than-proportional effect upon the engine thrust, as shown in Eq. (13.6). The inlet “ram recovery correction factor (C_{ram})” is provided by the manufacturer for various altitudes, Mach numbers, air temperatures, and thrust settings. Typically, C_{ram} ranges from 1.2–1.5. If the manufacturer’s data is not available, C_{ram} may be approximated as 1.35 for subsonic flight and by Eq. (13.7) for supersonic flight.

$$\text{Percent thrust loss} = C_{ram} \left[\left(\frac{P_1}{P_0} \right)_{ref} - \left(\frac{P_1}{P_0} \right)_{actual} \right] \times [100] \quad (13.6)$$

$$\text{Supersonic: } C_{ram} \cong 1.35 - 0.15(M_\infty - 1) \quad (13.7)$$

High-pressure air is bled from the engine compressor for cabin air, anti-icing, and other uses. This engine bleed air (not to be confused with inlet boundary-layer bleed and other forms of secondary airflow) exacts a thrust

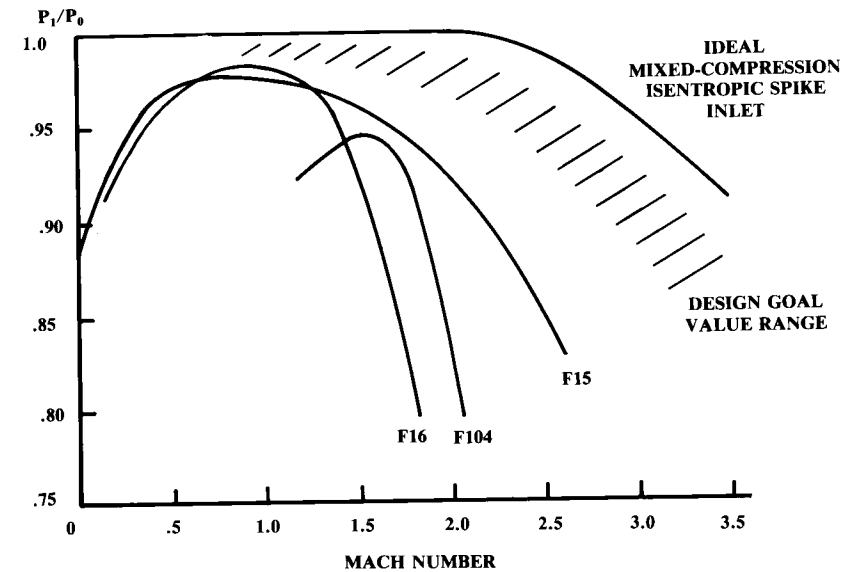


Fig. 13.4 Actual inlet pressure recoveries.

penalty that is also more-than-proportional to the percent of the total engine massflow extracted as bleed air.

Equation (13.8) illustrates this, where the “bleed correction factor (C_{bleed})” is provided by the manufacturer for various flight conditions. For initial analysis, C_{bleed} can be approximated as 2.0. The bleed massflow typically ranges from 1–5% of the engine mass-flow.

$$\text{Percent thrust loss} = C_{bleed} \left(\frac{\text{bleed mass flow}}{\text{engine mass flow}} \right) \times [100] \quad (13.8)$$

Installed engine thrust is also affected by horsepower extraction. Jet engines are equipped with rotating mechanical shafts turned by the turbines. The electrical generators, hydraulic pumps, and other such components connect to these shafts.

This extraction is typically less than 200 hp for a 30,000-lb-thrust engine, and usually has only a small effect upon installed thrust. Horsepower extraction is included in the cycle analysis used for detailed calculation of installed-engine thrust, but can be ignored for initial analysis.

As mentioned, moderate inlet distortion usually has little effect upon installed thrust, but can restrict the engine operating envelope. The effects of distortion are calculated later in the design process. For initial design, the

guidelines previously suggested for location of inlets and for forebody shaping should avoid any later problems with inlet distortion.

Nozzle efficiency has a direct effect upon thrust. However, it is rare to use a nozzle other than that provided by the manufacturer. For cases in which a unique nozzle (such as a 2-D vectoring nozzle) is employed, the new nozzle can usually be designed to provide the same efficiency as the manufacturer's nozzle. (The drag effects of alternate nozzles are discussed later.)

Installed Net Propulsive Force Corrections

The "installed engine thrust" is the actual thrust produced by the engine as installed in the aircraft. However, the engine creates three forms of drag that must be subtracted from the engine thrust to determine the thrust force actually available for propelling the aircraft. This propelling force, the "installed net propulsive force," is the thrust value to be used for aircraft performance calculations.

Most of the engine-related drag is produced by the inlet as a result of a mismatch between the amount of air demanded by the engine and the amount of air that the inlet can supply at a given flight condition. When the inlet is providing exactly the amount of air the engine demands (mass flow ratio equals 1.0), the inlet drag is negligible.

The inlet must be sized to provide enough air at the worst-case condition, when the engine demands a lot of air. This sets the capture area. Most of the time the engine demands less air than an inlet with this capture area would like to provide (i.e., mass flow ratio is less than 1.0).

When the mass flow ratio is less than 1.0, the excess air must either be spilled before the air enters the inlet or bypassed around the engine via a duct that dumps it overboard (Fig. 13.5) or into an ejector-type engine nozzle.

The drag from air spilled before entering the inlet is called "spillage," or "additive" drag. Additive drag represents a loss in momentum of the air which is slowed and compressed by the external part of the inlet but not used by the engine. The additive drag is determined by calculating, for each flight Mach number and engine mass-flow ratio, the Mach numbers and pressures throughout the inlet and integrating the forces in the flight direction for the part of the air which is spilled.

The spilled air will be turned back toward the freestream direction by the inlet cowl lip, producing a reduced pressure on the cowl. This provides a

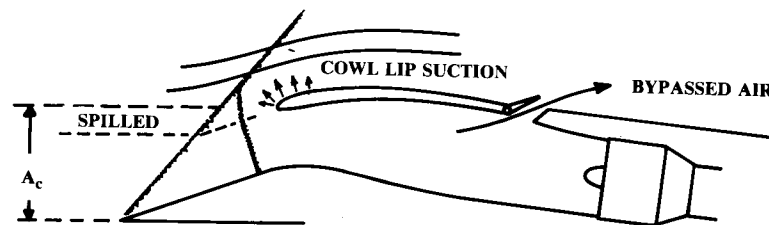


Fig. 13.5 Additive drag, cowl lip suction, and bypass subcritical operation.

suction with a component in the forward direction, i.e., a thrust (as shown in Fig. 13.5). This cowl-lip suction reduces the additive drag by as much as 30–40% in the low-supersonic regime. For a subsonic jet with well-rounded cowl lips, this suction will virtually eliminate additive drag.

Even with cowl-lip suction, the additive drag under certain flight conditions could exceed 20% of the total aircraft drag. A penalty of this magnitude is never seen because the designers resort to inlet-air bypass whenever the additive drag is too great.

Allowing the excess air to enter the inlet and be dumped overboard or into an ejector nozzle, will keep the inlet additive drag to a small value. The resulting bypass drag will be substantially less than the additive drag would have been. Bypass drag is calculated by summing the momentum loss experienced by the bypassed air.

Another form of inlet drag is the momentum loss associated with the inlet boundary-layer bleed. Air is bled through holes or slots on the inlet ramps and within the inlet to prevent shock-induced separation and to prevent the buildup of a thick turbulent boundary layer within the inlet duct. This air is dumped overboard out an aft-facing discharge exit, which is usually located a few feet behind the inlet.

(Note: don't confuse inlet boundary-layer bleed with the inlet boundary-layer diverter. The diverter prevents the fuselage boundary-layer air from entering the inlet. Diverter drag has been accounted for in the aerodynamics chapter.)

Calculation of bleed, bypass, and additive drag including cowl-lip suction is a complicated procedure combining analytical and empirical methods. The textbook methods (see Refs. 26, 27, 25, and 10) are very time-consuming and cannot account for the effects of the actual aircraft geometry, which may greatly affect both the inlet flowfield and the pressure loss through bleed and bypass ducts.

In a major aircraft company such calculations are made by propulsion specialists using complex computer programs. The results are included in the installed net propulsive force data that are provided to the sizing and performance analyst.

To permit rapid initial analysis and trade studies, Fig. 13.6 provides a "ballpark" estimate of inlet drag for a typical supersonic aircraft. This chart was prepared by the author using data from Ref. 47 and other sources, and should be used with great caution as it is merely typical data, not an estimate for any given inlet design.

This chart assumes that the engine is operating at a maximum dry or afterburning power setting, and that the inlet is operating at a corresponding mass-flow ratio. The chart does not reflect the increase in inlet drag experienced when the thrust setting is reduced (which reduces the mass-flow ratio). However, this chart should provide a reasonable approximation of inlet drag suitable for initial analysis and student design studies.

Nozzle drag varies with nozzle position as well as with the flight condition. To properly determine nozzle drag the actual nozzle geometry as a function of throttle setting and flight condition must be known, and the drag calculated by taking into account the overall aircraft flowfield. As an initial approximation, the effect of nozzle position may be ignored and the

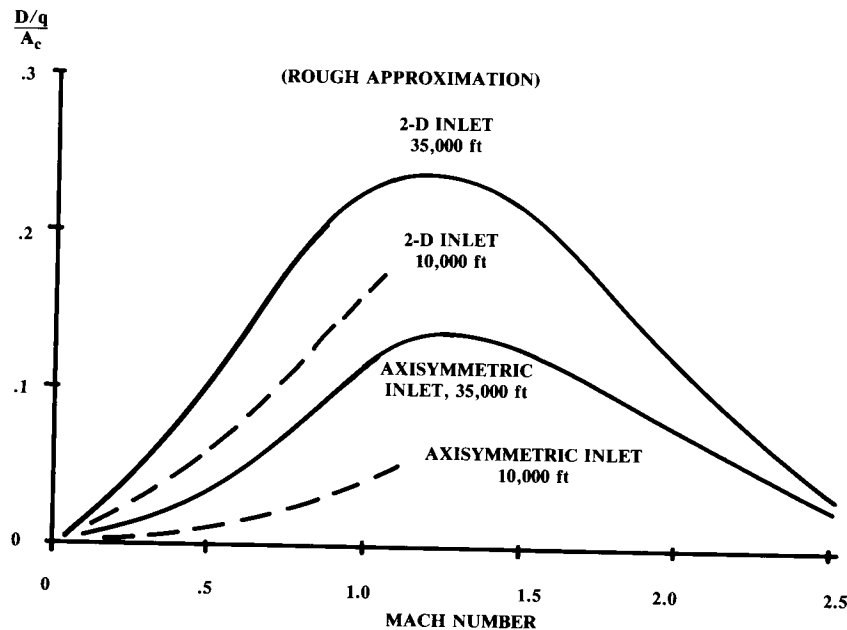


Fig. 13.6 Inlet drag trends.

nozzle drag estimated by the typical subsonic values shown in Table 13.1 (Ref. 10) for the nozzle types shown in Fig. 10.20.

The nozzle drag increases transonically and then drops off at supersonic speeds. For initial analysis the subsonic value may be assumed for all speeds. Note that these nozzle drags are referenced to the maximum cross-sectional area of the fuselage. For a subsonic, podded nacelle, the nozzle drag is negligible.

The remaining propulsion-system drag is the variation of trim drag with throttle setting. If the engine thrust axis is not through the center of gravity, any thrust change will cause a pitching moment. The trim force required to counter this moment is charged to the propulsion in most thrust-drag book-

Table 13.1 Nozzle incremental drag¹⁰

Nozzle type	Subsonic $\Delta C_{D_{Fuselage}}$ *
Convergent	.036-.042
Convergent iris	.001-.020
Ejector	.025-.035
Variable ejector	.010-.020
Translating plug	.015-.020
2-D nozzle	.005-.015

*Referenced to fuselage maximum cross-section area.

keeping systems. For initial analysis this may be ignored unless the thrust line is substantially above or below the aircraft centerline.

13.6 PISTON-ENGINE PERFORMANCE

The aircraft piston engine operates on the four-stroke Otto cycle used by automobiles. The thermodynamic theory of the Otto-cycle reciprocating engine is described in Refs. 28, 48, and 49. For design purposes the most important thing to know about the piston engine is that the horsepower produced is directly proportional to the massflow of the air into the intake manifold. In fact, horsepower is approximately 620 times the air massflow (lb/s).

Mass flow into the engine is affected by the outside air density (altitude, temperature, and humidity) and intake manifold pressure. Equation (13.9) accounts for the air-density effect upon horsepower, and is attributed to Gagg and Ferrar of the Wright Aeronautical Company (1934). This equation indicates that at an altitude of 20,000 ft a piston engine has less than half of its sea-level horsepower.

$$bhp = bhp_{SL} \left(\frac{\rho}{\rho_0} - \frac{1 - \rho/\rho_0}{7.55} \right) \quad (13.9)$$

The intake manifold is usually at atmospheric pressure. A forward-facing air-intake scoop can provide some small increase in manifold pressure at higher speeds. Large increases in manifold pressure require mechanical pumping via a "supercharger" or "turbosupercharger."

The supercharger is a centrifugal air compressor mechanically driven by a shaft from the engine. The amount of air compression available is proportional to engine RPM. The turbosupercharger, or "turbocharger," is driven by a turbine placed in the exhaust pipe. This recovers energy which would otherwise be wasted, and decouples the available amount of compression from the engine RPM.

Supercharging or turbocharging is usually used to maintain sea level pressure in the intake manifold as the aircraft climbs. Typically the sea-level pressure can be maintained up to an altitude of about 15,000–20,000 ft. Above this altitude the manifold pressure, and hence the horsepower, drops. Figure 13.7 shows typical engine performance for nonsupercharged, supercharged, and turbocharged engines.

Supercharging or turbocharging may also be used to raise the intake manifold pressure above the sea-level value to provide additional horsepower from a given engine. However, the increased internal pressures require a heavier engine for structural reasons.

Piston engine performance charts are provided by the manufacturer as a function of manifold pressure, altitude, and RPM.

Propeller Performance

A propeller is a rotating airfoil that generates thrust much as a wing generates lift. Like a wing, the propeller is designed to a particular flight

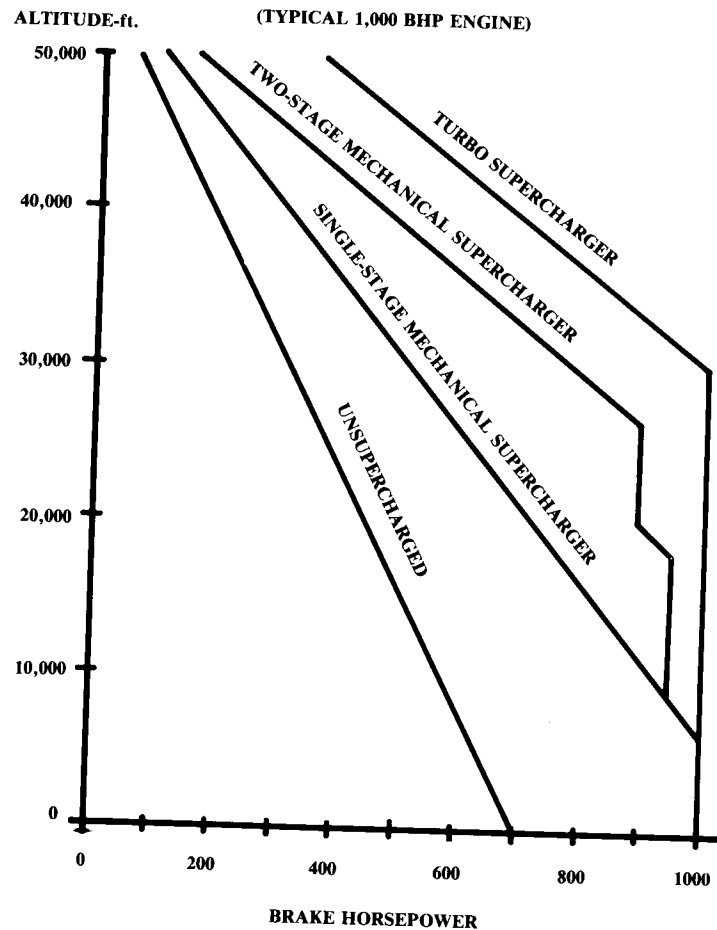


Fig. 13.7 Effects of supercharging.

condition. The propeller airfoil has a selected design lift coefficient (usually around 0.5), and the twist of the airfoil is selected to give the optimal airfoil angle of attack at the design condition.

Since the tangential velocities of the propeller airfoil sections increase with distance from the hub, the airfoils must be set at progressively reduced pitch-angles going from root to tip. The overall “pitch” of a propeller refers to the blade angle at 75% of the radius (70% in some books).

Propeller theory is well covered in many textbooks, such as Ref. 49. While theory is useful for propeller designers, the aircraft designers usually work with experimental propeller data provided by the propeller companies.

As for a wing, the properties of a propeller are expressed in coefficient form. Experimental data for design purposes are expressed using a variety of parameters and coefficients, as described below.

$$\text{Advance Ratio: } J = V/nD \quad (13.10)$$

$$\text{Power Coefficient: } c_P = \frac{P}{\rho n^3 D^5} = \frac{550 \text{ bhp}}{\rho n^3 D^5} \quad (13.11)$$

$$\text{Thrust Coefficient: } c_T = T/\rho n^2 D^4 \quad (13.12)$$

$$\text{Speed-Power Coefficient: } c_S = V \sqrt[5]{\rho/Pn^2} \quad (13.13)$$

Activity Factor:

$$AF_{\text{per blade}} = \frac{10^5}{D^5} \int_{0.15R}^R cr^3 dr = \frac{10^5 c_{\text{root}}}{16D} [0.25 - (1 - \lambda)0.2] \quad (13.14)$$

$$\text{Propeller Efficiency: } \eta_p = \frac{TV}{P} = \frac{TV}{550 \text{ bhp}} = \frac{J c_T}{c_P} \quad (13.15)$$

$$\text{Thrust: } T = \frac{550 \text{ bhp } \eta_p}{V} = \frac{c_T}{c_P} \frac{550 \text{ bhp}}{nD} \quad (13.16)$$

where

- T = thrust (lb)
- V = velocity (ft/s)
- P = power (ft-lb/s)
- bhp = brake horsepower
- n = rotation speed (rev/s)
- D = propeller diameter (ft)
- c = propeller airfoil chord (ft)

The advance ratio (equivalent to the wing angle of attack) is related to the distance the aircraft moves with one turn of the propeller. Advance ratio is sometimes called the “slip function” or “progression factor.”

The power and thrust coefficients are nondimensional measures of those quantities, much like the wing lift-coefficient. The speed-power coefficient is defined as the advance ratio raised to the fifth power divided by the power coefficient. The speed-power coefficient is nondimensional and does not involve the propeller diameter, which is useful for comparison between propellers of different size.

The activity factor is a measure of the amount of power being absorbed by the propeller. Activity factors range from about 90–200, with a typical light-aircraft activity factor being 100 and a typical large turboprop having an activity factor of 140. The final expression in Eq. (13.14) is the activity factor for a straight-tapered propeller blade of taper ratio λ .

Equation (13.15) relates the propeller efficiency, previously discussed in Chapters 3 and 10, to the advance ratio and the ratio of the thrust coefficient to the power coefficient. This ratio is used in Eq. (13.16) to determine the thrust at static conditions when the velocity is zero and the propeller-efficiency equation cannot be used for thrust determination.

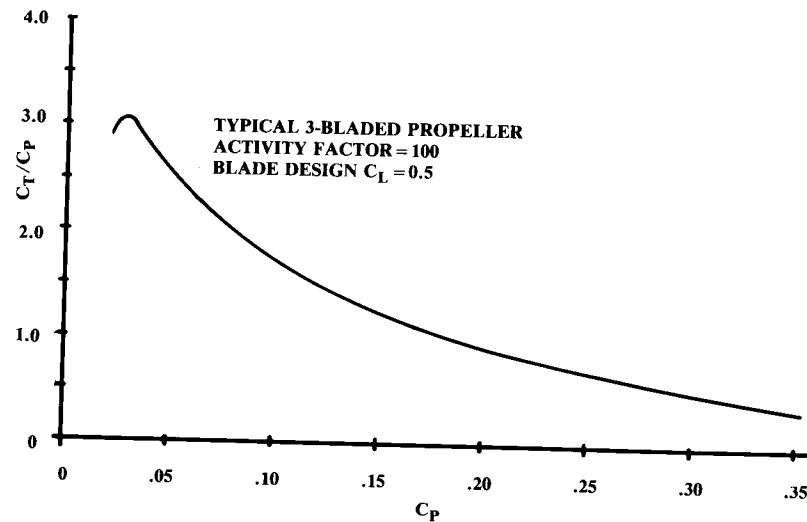


Fig. 13.8 Static propeller thrust. (after Ref. 50)

Propeller data is available from the manufacturers as well as various NASA/NACA reports. This data is provided in many different formats using different combinations of the above parameters and coefficients. Whatever the format, Eq. (13.16) is ultimately used to determine the propeller thrust at a given flight condition.

Figures 13.8 and 13.9, propeller charts for static and forward flight (Ref. 50), have been chosen as typical of propellers used for modern light and business aircraft. These charts represent a three-bladed propeller with a design lift-coefficient of 0.5 and an activity factor of 100.

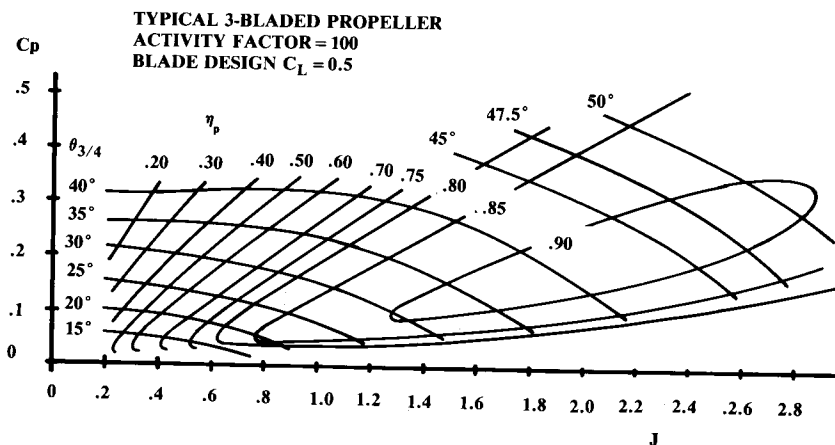


Fig. 13.9 Forward flight thrust and efficiency. (after Ref. 50)

For a two-bladed propeller, the forward-flight efficiencies are about 3% better than shown in Fig. 13.9, but the static thrust is about 5% less than shown in Fig. 13.8. The reverse trends are true for a four-bladed propeller. Also, a wooden propeller has an efficiency about 10% lower due to its greater thickness.

If Ref. 50 is unavailable, Ref. 49 contains 43 pages of propeller charts taken from Ref. 50. Reference 49 also contains a compressibility correction for a propeller with a high tip Mach number (greater than 0.9).

If the propeller is of variable-pitch design, its pitch is adjusted to the optimum blade angle at each flight condition to produce a constant engine RPM regardless of the horsepower being produced.

The advance ratio and power coefficient are then independent variables and the propeller efficiency can be read in Fig. 13.9 for any combination of advance ratio and power coefficient that may occur in flight. Blade angle for the variable-pitch propeller can be read as a fallout parameter in Fig. 13.9.

Remember that the propeller thrust in forward flight is proportional to the inverse of the velocity, which would imply infinite thrust at zero velocity. Instead, the propeller produces the static-thrust value from Fig. 13.8 at zero velocity.

In the speed range from zero to about 50 knots (such as during takeoff), the thrust varies in a fashion that can be represented by a smooth curve faired between the static-thrust value and the calculated forward-flight thrust.

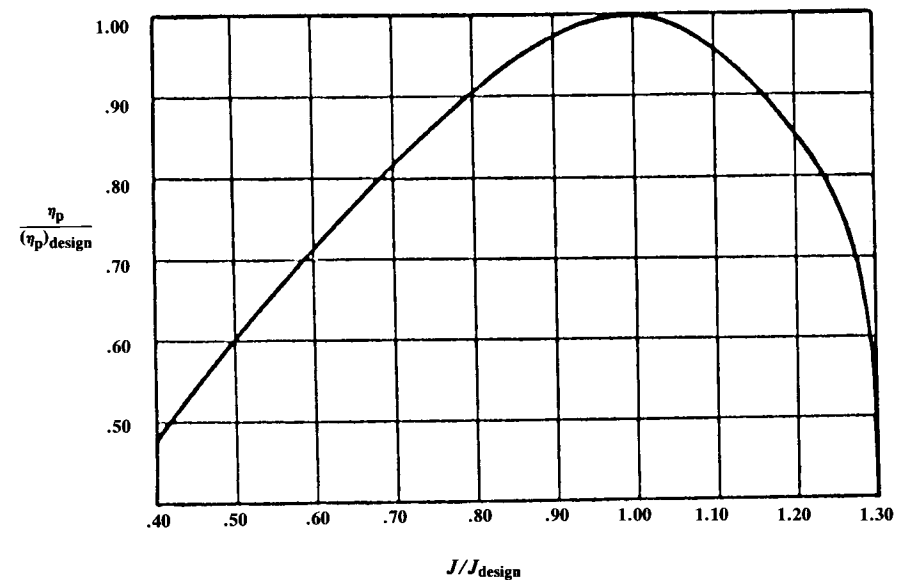


Fig. 13.10 Fixed-pitch propeller adjustment.

If a fixed-pitch propeller is used, the blade angle cannot be varied in flight to maintain engine RPM at any flight condition. Since the RPM and therefore horsepower will vary with velocity, the efficiency and hence the thrust will be reduced at any speed other than the design speed.

Figure 13.9 could be used to determine the thrust from a fixed-pitch propeller by following the appropriate line for the selected blade angle. However, it is simpler to use the approximate method of Fig. 13.10 unless actual propeller data is available.

Figure 13.10 relates the fixed-pitch propeller efficiency at an off-design velocity and RPM to the on-design efficiency, which is attained by the propeller at some selected flight condition. The on-design efficiency is obtained from Fig. 13.9, which is also used to get the required blade angle for the design condition.

The static thrust of a fixed-pitch propeller will be less than is estimated using Fig. 13.8. A fixed-pitch propeller suffers at low speeds due to the high local angles of attack of the blades at low speeds and high RPMs. As a rough approximation it can be assumed that the static thrust is about equal to the thrust at 50 knots for the fixed-pitch propeller.

These charts provide useful rough estimations of propeller performance, but actual charts for the selected propeller should be obtained from the manufacturer for any serious design effort.

Piston-Prop Thrust Corrections

As with jet engines, there are several engine-related drag items that must be considered, namely, scrubbing drag, cooling drag, and engine miscellaneous drag.

Scrubbing drag is the increase in aircraft drag due to the higher velocity and turbulence experienced by the parts of the aircraft within the propwash. This drag could be calculated by determining, for each flight condition, the increased dynamic pressure within the propwash and using that value for the component-drag calculation.

A simpler approach, called the SBAC (Society of British Aircraft Constructors) method, adjusts the propeller efficiency as in Eq. (13.17). The subscript “washed” refers to the parts of the aircraft which lie within the propwash. If the parasite-drag coefficient for the propwashed parts of the aircraft cannot be determined, 0.004 is a reasonable estimate.

$$\eta_{p\text{effective}} = \eta_p \left[1 - \frac{1.558}{D^2} \frac{\rho}{\rho_0} \Sigma(C_{f_e} S_{\text{wet}})_{\text{washed}} \right] \quad (13.17)$$

where C_{f_e} is the equivalent skin-friction (parasite) drag coefficient, referenced to wetted area.

For a pusher-propeller configuration, the scrubbing drag is zero. However, the pusher propeller suffers a loss of efficiency due to the wake of the fuselage and wing. This loss is strongly affected by the actual aircraft configuration, and should equal about 2–5%.

Cooling drag represents the momentum loss of the air passed over the engine for cooling. This is highly dependent upon the detail design of the intake, baffles, and exit.

Miscellaneous engine drag includes the drag of the oil cooler, air intake, exhaust pipes, and other parts. Cooling and miscellaneous drags for a well-designed engine installation can be estimated by Eqs. (13.18) and (13.19) (Ref. 23). However, a typical light aircraft engine installation may experience cooling and miscellaneous drag levels 2–3 times the values estimated by these equations.

$$(D/q)_{\text{cooling}} = (4.9 \times 10^{-7}) \frac{\text{bhp} \cdot T^2}{\sigma V}, \text{ ft}^2 \quad (13.18)$$

$$(D/q)_{\text{misc}} = (2 \times 10^{-4}) \text{ bhp}, \text{ ft}^2 \quad (13.19)$$

where

T = air temperature, deg Rankine

V = velocity in ft/s

13.7 TURBOPROP PERFORMANCE

A turboprop is a jet engine that drives a propeller using a turbine in the exhaust. The jet exhaust retains some thrust capability, and can contribute as much as 20% of the total thrust. For this reason the horsepower rating of a turboprop engine includes the horsepower equivalent of this residual thrust.

This horsepower equivalent of residual thrust is arbitrarily calculated under static conditions as the residual thrust divided by 2.5. Under forward-flight conditions it is calculated using Eq. (13.16) assuming that the propeller efficiency $\eta_p = 0.80$. The total of the mechanical and thrust residual horsepower is called the “equivalent shaft horsepower (ESHP).”

Analysis of the turboprop is a hybrid between the jet and the piston-prop analysis. The engine is analyzed like a jet, including the inlet effects. The residual thrust is provided by the manufacturer as a horsepower equivalent. The propeller is analyzed as described above, including the scrubbing-drag term.

The conventional turboprop, like the piston-prop, is limited by tip Mach number to about Mach 0.7. The turboprop has higher efficiency than the piston-prop at Mach numbers greater than about 0.5 due to the residual jet thrust, but the conventional turboprop is no match for a turbofan engine at the higher subsonic speeds.

Recently, a new type of advanced propeller has been developed that offers good efficiencies up to about Mach 0.85. These are known as “propfans” or “unducted fans (UDF).” They are smaller in diameter than the regular propellers and feature numerous wide, thin, and swept blades. Test programs to date indicate that a well-designed propfan can retain propeller efficiencies of over 0.8 at speeds on the order of Mach 0.85.

Up-to-date data on propfans can be found in publications such as the AIAA’s *Journal of Aircraft* and *Journal of Propulsion and Power*.